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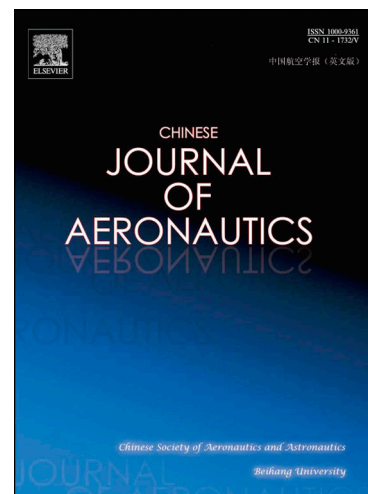
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A turbulence-chemistry recursive mixing model for TPDF simulations and its appraisal

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Abstract The closure of micro-mixing processes constitutes a critical challenge for Transported Probability Density Function (TPDF) methods to achieve high accuracy and universality in high-fidelity combustor simulations. This study proposes a Turbulence-Chemistry Recursive (TCR) mixing model that eliminates adjustable model constants through autonomous relaxation toward a turbulence-chemistry manifold. Grounded in turbulence-chemistry interactions that drive micro-mixing toward dynamic equilibrium rather than complete homogenization, TCR establishes a self-consistent relationship between filtered TPDF reaction rates and perfectly stirred reactor references. This enables inverse determination of instantaneous manifold states, achieving closure via recursive integration of physical constraints into TPDF solutions. Rigorous validation in turbulent lifted flames demonstrates TCR's superior predictive accuracy over Interaction by Exchange with the Mean (IEM) in temperature and species concentration statistics, particularly in flame base regions, while maintaining comparable computational costs. By coupling turbulence-driven homogenization with chemistry-driven differentiation through real-time target updates, the model achieves autonomous adaptation across Damköhler regimes. This approach establishes TCR as a critical advancement that effectively enhances the Eulerian stochastic field method for TPDF simulations, enabling high-accuracy predictions across wide operating conditions for engineering applications.

1. Introduction

In recent years, combustor design has transitioned from conventional approaches focusing on macroscopic characteristics to refined methodologies driven by precision engineering demands, necessitating high-fidelity simulations¹⁻⁵. The closure of chemi-

cal reaction source terms remains pivotal for combustion modeling^{6,7}, where Large-Eddy Simulation (LES) coupled with Transported Probability Density Function (TPDF) methodology provides an assumption-free closure framework for turbulence-chemistry interactions⁸. The LES-TPDF framework achieves the closure of the term of the Subgrid Scale (SGS) chemical source without presuming low-dimensional manifolds, allowing theoretical applicability to both premixed and non-premixed combustion regimes^{9,10}. This generality allows high-accuracy predictions of combustors operating under environmentally constrained and highly transient conditions. Despite exact closure of chemical source terms, the micro-mixing term in TPDF methods still requires accurate modeling^{11,12}. Molecular mixing critically determines the species distributions, posing a central challenge for the closure of the TPDF under strong

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turbulence-chemistry interaction^{13,14}.

To address the micro-mixing closure in TPDF methods, various SGS mixing models have been developed, including the Interaction by Exchange with the Mean (IEM) model¹⁵, Modified Curl (MC) model^{16,17}, and Euclidean Minimum Spanning Tree (EMST) model¹⁸. Although these models achieve satisfactory accuracy in certain combustion test cases, they have long been criticized for underrepresenting the influence of chemical reactions on molecular mixing, a critical limitation in combustor simulation where turbulence-chemistry interactions span multiple spatial and temporal scales, particularly evident in simple formulations like the IEM and MC. Although the EMST model partially captures the spatial locality of the effects of the turbulence-chemistry interaction on molecular mixing by introducing composition-space locality, it violates the principles of linearity and independence while generating artificial chainlike structures in the composition space.

The turbulence-chemistry interaction exhibits localized effects on micro-mixing, prompting the development of advanced models like Multiple Mapping Conditioning (MMC)^{19,20}. These approaches introduce auxiliary reference variables as conditional constraints for micro-mixing, enabling flexible adjustment of locality while ensuring relaxation to Gaussian probability density functions. Other theoretically rigorous and complex mixing model approaches include the Fokker-Planck closure models²¹, where the requirement to specify the joint conditional scalar dissipation rate introduces a standalone challenge in model formulation. Although surpassing conventional models in adaptability, their computational complexity significantly exceeds that of IEM and MC models. New mixing models can be developed by introducing additional parameters to interpolate mixing behavior between classical model characteristics, as exemplified by the Shadow Position Mixing Model (SPMM)^{22,23}, which achieves EMST-like or IEM-like mixing through parametric adjustments though incurs critical computational challenges in three-dimensional flows due to three-variable conditional mean evaluations. An alternative, the pairwise mixing model with Kernel constraint (KerM)²⁴, balances locality and computational efficiency by regulating the kernel size to achieve a micro-mixing behavior intermediate between the EMST and MC models. While these novel models demonstrate potential advantages through tunable locality, their performance exhibits high sensitivity to parameter selection, necessitating case-specific calibrations^{25–27}. This dependency imposes critical challenges in parameter universality and practical applicability across diverse combustion scenarios.

In existing micro-mixing models, beyond the parameters introduced in specific formulations to enhance locality adjustment, a unified model parameter C_ϕ representing the mechanical-to-scalar time-scale governs the mixing frequency. Previous studies demonstrate that TPDF calculations exhibit a significant sensitivity to C_ϕ values²⁸, with optimal C_ϕ magnitudes that span an order of magnitude (1.0–3.0¹¹, 1.5–3.8²⁹, 5.0³⁰, 8.0³¹, and 10.0³²) across different combustor configurations reported in the literature. To locally adapt C_ϕ in combustion fields, one approach introduces empirical relationships between mixing and chemical timescales for dynamic adjustment³³. However, this method lacks physical foundation due to the absence of rigorous derivation for the timescale formulations. Kuron et al.³⁴ developed a dynamic mixing model for premixed flames by partitioning SGS scalar mixing via a segregation factor into two regimes: turbulence-dominated mixing with constant mechanical-to-scalar time-scale and laminar flamelet-dominated mixing governed by diffusion/chemistry interactions, achieving smooth transitional blending through TPDF-resolvable parameters that adaptively bridge turbulence-induced mixing and

laminar flamelets combustion physics^{35,36}. An alternative dynamic procedure determines C_ϕ via test-filtered SGS quantities^{37,38}, requiring the coefficient to behave as a spatially/temporally slowly varying statistical quantity with scale invariance and directional homogeneity³⁹. While applicable to regions with large turbulence-chemistry interaction scales, this approach fails near flame fronts where chemical timescales are short. Here, C_ϕ may exhibit sharp spatial variations exceeding the test filter's effective resolution, potentially inducing nonphysical fluctuations in model predictions^{40,41}.

The conceptual framework for addressing turbulence-chemistry interactions in micro-mixing closure bears notable parallels with the Partially Stirred Reactor (PaSR) modeling paradigm^{42,43}. In PaSR formulations, the SGS mixing process is conceptualized as a dynamic equilibrium between turbulent homogenization and chemical segregation, where turbulence drives scalar fields toward uniformity while finite-rate chemistry counteracts this tendency. This dual mechanism partitions computational cells into two conceptual zones: a Perfectly Stirred Reactor (PSR) zone governed by chemical timescales and an unmixed reactant reservoir regulated by turbulent mixing timescales. The filtered reaction rate is subsequently derived through timescale competition between these counteracting processes. While demonstrating reliable accuracy in simulating various combustors^{44–46}, the conventional PaSR model exhibits limitations in handling multi-scalar mixing processes⁴⁷ and suffers from ambiguous definitions of turbulent mixing and chemical timescale formulations^{48–51}. While differing in implementation details from TPDF micro-mixing closures, the PaSR approach shares the fundamental premise of dynamically coupling turbulence and chemistry timescales to determine SGS reaction rates, providing valuable insights for developing physics-constrained mixing models without empirical constants.

In this work, a Turbulence-Chemistry Recursive (TCR) mixing model is proposed and its performance is evaluated. The new model builds upon the physical insight, i.e., turbulence-chemistry interactions drive micro-mixing toward dynamic equilibrium states rather than complete homogenization. By coupling filtered reaction rates derived from TPDF solutions with an enhanced multi-layer PaSR framework, the TCR model dynamically determines equilibrium mixing states. As the Damköhler number (Da) approaches zero, it converges to the IEM model with test-filtered determination of C_ϕ , while demonstrating reduced micro-mixing intensity at elevated Da . The TCR model demonstrates fundamental improvements over conventional dynamic mixing closures through two key innovations: elimination of empirical parameters by fixing C_ϕ at its test-filtered value under low- Da conditions while preserving localized responsiveness to micro-mixing characteristics, and physically constrained implementation of compartmentalization concept through rigorous TPDF-PSR reaction rate correlations rather than phenomenological timescale assumptions. This recursive framework establishes a closed-loop system where micro-mixing modeling parameters are intrinsically determined through TPDF solutions of filtered reaction rates, which subsequently feed back into the mixing operator. The proposed TCR approach also provides a novel alternative for Eulerian Stochastic Field (ESF) implementations, as illustrated in Fig. 1, particularly valuable given that most micro-mixing models rely on Lagrangian particle data despite the practical advantages of ESF methods for TPDF simulations.

To evaluate the performance of the proposed TCR mixing model, ESF-TPDF simulations of turbulent lifted flames are conducted within an extensively validated software framework^{5,33,52–54} and compared against experimental data and simulation results obtained using the conventional IEM model. The rest of this paper is structured as follows: Section 2 details the ESF-TPDF method

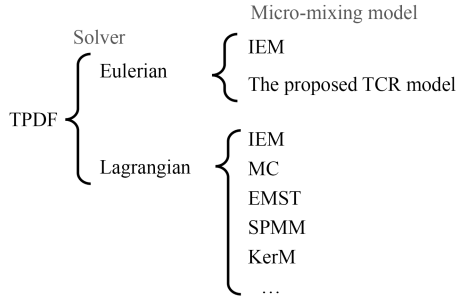


Fig. 1 Schematics of micro-mixing models and their implementation in TPDF, contrasting Lagrangian particle and Eulerian stochastic field frameworks.

and the TCR mixing model. Section 3 describes the computational configurations and numerical settings for the turbulent lifted flame simulations. Section 4 presents comparative analyses of simulation results based on IEM and TCR mixing model with experimental data.

2. Methodology

2.1. ESF-TPDF method

The TPDF method within the LES framework provides a rigorous mathematical approach for turbulent combustion modeling through solution of filtered Probability Density Function (PDF) transport equations. The SGS joint PDF is defined as⁵⁵:

$$P_{\text{SGS}}(\psi; \mathbf{x}, t) = \int_V \mathcal{F}(\psi; \mathbf{x}', t) G(\mathbf{x} - \mathbf{x}', \Delta) d\mathbf{x}' \quad (1)$$

where V denotes the control volume and G represents the LES filter kernel with characteristic width Δ .

The fine-grained PDF $\mathcal{F}(\psi; \mathbf{x}, t)$ is expressed through Dirac delta functions as $\mathcal{F}(\psi; \mathbf{x}, t) = \prod_{\alpha=1}^{N_{\text{sc}}} \delta(\psi_{\alpha} - \phi_{\alpha}(\mathbf{x}, t))$. The composition vector $\psi = [\psi_1, \psi_2, \dots, \psi_{N_s}, \psi_h]$ spans the sample space corresponding to the physical scalar vector $\phi = [Y_1, Y_2, \dots, Y_{N_s}, h]$, where N_s indicates the number of chemical species, $N_{\text{sc}} = N_s + 1$ represents the total scalar count, Y_{α} the mass fraction of the α -th species (where α ranges over all species), and h enthalpy. The SGS joint PDF $P_{\text{SGS}}(\psi; \mathbf{x}, t)$ statistically quantifies the probability density of the thermochemical state $\phi(\mathbf{x}, t)$ attaining the composition-space value ψ . Its transport equation is derived as^{56,57}:

$$\begin{aligned} \bar{\rho} \frac{\partial P_{\text{SGS}}}{\partial t} + \bar{\rho} \tilde{u}_j \frac{\partial P_{\text{SGS}}}{\partial x_j} - \sum_{\alpha=1}^{N_{\text{sc}}} \frac{\partial}{\partial \psi_{\alpha}} (\rho \dot{\omega}_{\alpha} P_{\text{SGS}}) = \\ - \frac{\partial}{\partial x_j} \left(\overline{\rho \mathcal{F} u_j} - \bar{\rho} \tilde{u}_j P_{\text{SGS}} \right) - \sum_{\alpha=1}^{N_{\text{sc}}} \frac{\partial}{\partial \psi_{\alpha}} \left(\frac{\mu}{\sigma_{\alpha}} \cdot \frac{\partial \phi_{\alpha}}{\partial x_i \partial x_i} \mathcal{F} \right) \end{aligned} \quad (2)$$

where $N_{\text{sc}} = N_s + 1$, u_j denotes the velocity component in the j -th direction, $\dot{\omega}_{\alpha}$ denotes the reaction rate of scalar α , μ represents the molecular viscosity, σ_{α} is the Schmidt number for scalar α incorporating differential diffusion effect⁵⁴, the overbar $\bar{\cdot}$ denotes conventional filtered quantities, and the tilde $\tilde{\cdot}$ indicates Favre-filtered variables.

The first term on the right-hand side of Eq. (2) is closed through a gradient model as:

$$- \frac{\partial}{\partial x_j} \left(\overline{\rho \mathcal{F} u_j} - \bar{\rho} \tilde{u}_j P_{\text{SGS}} \right) = \frac{\partial}{\partial x_j} \left(\frac{\mu_{\text{SGS}}}{\sigma_{\text{SGS}}} \cdot \frac{\partial P_{\text{SGS}}}{\partial x_j} \right) \quad (3)$$

where the SGS viscosity μ_{SGS} is obtained from the turbulence model and the turbulent Schmidt number σ_{SGS} takes the optimal value 0.4 as recommended by LES studies^{58–60}.

The second term on the right-hand side of Eq. (2) represents the filtered conditional scalar dissipation rate, which quantifies the SGS mixing effects and describes the influence of molecular diffusion processes on the evolution of P_{SGS} . In the conventional IEM model, this term is closed as:

$$\begin{aligned} - \sum_{\alpha=1}^{N_{\text{sc}}} \frac{\partial}{\partial \psi_{\alpha}} \left(\frac{\mu}{\sigma_{\alpha}} \cdot \frac{\partial \phi_{\alpha}}{\partial x_i \partial x_i} \mathcal{F} \right) = \\ - \frac{1}{\tau_{\text{SGS}}} \sum_{\alpha=1}^{N_{\text{sc}}} \frac{\partial}{\partial \psi_{\alpha}} \left[(\psi_{\alpha} - \tilde{\phi}_{\alpha}) \bar{\rho} P_{\text{SGS}} \right] \end{aligned} \quad (4)$$

where τ_{SGS} denotes the SGS mixing timescale⁶¹.

The ESF method⁶² solves the resulting system through an ensemble of stochastic differential equations. In this approach, P_{SGS} is represented by N stochastic fields for each of the N_{sc} scalars, denoted as $\xi_{\alpha}^n(\mathbf{x}, t)$ where $1 \leq n \leq N$. Adopting the Itô interpretation for stochastic integration, the transport equation for each stochastic field becomes:

$$\begin{aligned} \bar{\rho} d\xi_{\alpha}^n = - \bar{\rho} \tilde{u}_i \frac{\partial \xi_{\alpha}^n}{\partial x_i} dt + \frac{\partial}{\partial x_i} \left(\Gamma_{\alpha} \frac{\partial \xi_{\alpha}^n}{\partial x_i} \right) dt \\ + \sqrt{2 \bar{\rho} \Gamma_{\alpha}} \frac{\partial \xi_{\alpha}^n}{\partial x_i} dW_i^n + \mathcal{M} + \bar{\rho} \dot{\omega}_{\alpha}^n dt \end{aligned} \quad (5)$$

where $\Gamma_{\alpha} = \mu/\sigma_{\alpha} + \mu_{\text{SGS}}/\sigma_{\text{SGS}}$ denotes the total diffusion coefficient⁵⁴ of α . The Wiener process increment dW_i^n corresponds to a vector-valued stochastic forcing term with independent components in each spatial direction (i) and stochastic field (n). $\phi_{\alpha} = \frac{1}{N} \sum_{n=1}^N \xi_{\alpha}^n$ and $\dot{\omega}_{\alpha}(\phi) = \frac{1}{N} \sum_{n=1}^N \dot{\omega}_{\alpha}^n(\xi^n)$.

In Eq. (5), the micro-mixing term implements the IEM closure through:

$$\mathcal{M} = - \frac{\bar{\rho}}{2\tau_{\text{SGS}}} \left(\xi_{\alpha}^n - \tilde{\phi}_{\alpha} \right) dt \quad (6)$$

The SGS turbulent mixing timescale τ_{SGS} is conventionally modeled as proportional to the characteristic turbulence timescale through the relationship $\tau_{\text{SGS}} = \tau_{\text{Turb}}/C_{\phi}$, where C_{ϕ} represents a model constant. The determination of τ_{Turb} depends fundamentally on the employed turbulence modeling framework: in Reynolds-Averaged Navier-Stokes (RANS) formulations, it is typically defined as the ratio of turbulent kinetic energy to dissipation rate (k/ε)²⁸, whereas in LES/ESF-TPDF approaches, it is generally expressed as $\bar{\rho} \Delta^2 / (\mu + \mu_{\text{SGS}})$ ⁶³, incorporating the filtered density $\bar{\rho}$, filter width Δ , molecular viscosity μ , and SGS turbulent viscosity μ_{SGS} .

The model coefficient C_{ϕ} is conventionally assigned a fixed value typically ranging from 1.5 to 2.5 in non-reacting shear flows⁶⁴. In LES/TPDF simulations, where C_{ϕ} is frequently denoted as C_M or C_d , its optimal value exhibits case-dependent variation, e.g., approximately 1.5–2.0 for Sandia jet flames⁶⁵, 2.0–10.0 for Sandia/Sydney flame SMA1³², and 5.0 for H_2/CO non-premixed flames³⁰. The parameter generally operates within the 1.0–10.0 range across LES/TPDF applications, with values exceeding 10.0 yielding negligible improvements as SGS scalar distributions approach homogeneity³⁸. Building on our prior LES/ESF-TPDF studies employing the IEM model with $C_{\phi} = 2.0$, which successfully reproduced experimental data for diverse configurations including jet flames^{33,66}, bluff-body stabilized flames^{52,67}, and practical combustors^{5,68}, this paper adopts the $C_{\phi} = 2.0$ IEM implementation as the baseline comparator for evaluating the new mixing model.

2.2. TCR mixing model

The TCR mixing model introduces a shift in micro-mixing closure by driving scalar fields toward turbulence-chemistry equilibrium states rather than enforcing complete homogeneity. While maintaining the mathematical framework of the IEM model, TCR replaces the conventional homogenization target $\tilde{\phi}_\alpha$ with a dynamically evolving equilibrium state ψ_α^n . This equilibrium state is determined through coupled analysis of filtered reaction rates from TPDF solutions and reference PSR calculations. The resulting micro-mixing term in Eq. (5) becomes:

$$\mathcal{M} = -\frac{\bar{\rho}}{2\tau_{\text{SGS}}^{\text{TCR}}} (\xi_\alpha^n - \psi_\alpha^n) \text{d}t \quad (7)$$

where scalar compositions ξ_α^n evolve toward the equilibrium state ψ_α^n at a rate governed by the timescale $\tau_{\text{SGS}}^{\text{TCR}}$.

The micro-mixing term expression defines the rate of change of stochastic variables as $\text{d}\xi_\alpha^n/\text{d}t = \mathcal{M}/\text{d}t$. After undergoing Wiener stochastic transport and micro-mixing processes, the stochastic variable ξ_α^n ideally converges toward data at randomly distributed positions within a grid cell, as is shown in Fig. 2. The background color in the figure illustrates the distribution of scalar ϕ_α . Each ξ_α^n represents a random spatial sample within the grid cell. As the sample size approaches infinity ($N \rightarrow \infty$), the sampling locations uniformly populate the control volume. The ensemble of stochastic variables consequently reflects the complete set of ϕ_α values at all positions throughout the control volume.

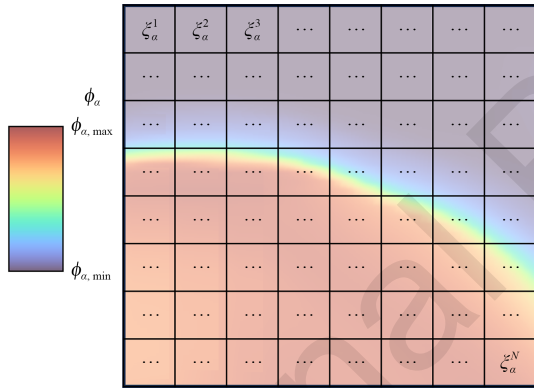


Fig. 2 Schematic diagram of ideal stochastic variable values after random transport and micro-mixing.

Consequently, for identical mean values, greater non-uniformity in the grid's scalar distribution yields proportionally larger post-transport/mixing variance in ξ_α^n . This relationship is implemented in the TCR model through ψ_α^n , which interpolates between ξ_α^n and $\tilde{\phi}_\alpha$ based on local transient turbulence-chemistry interactions: $\psi_\alpha^n = \xi_\alpha^n$ signifies extreme scalar inhomogeneity analogous to wrinkled flamelet, whereas $\psi_\alpha^n = \tilde{\phi}_\alpha$ indicates homogeneous scalar distributions characteristic of distributed reaction zones. By partitioning computational cells based on local transient chemical inhomogeneity, the TCR model divides each cell into a perfectly stirred fraction (where scalars attain the mean value) and a completely unmixed fraction (where scalar variance remains unaltered).

2.2.1. Formulation of SGS mixing frequency

To dynamically obtain the model parameter C_ϕ in the mixing frequency, commonly used models for conserved scalar variance and dissipation rate are employed⁶⁹:

$$\widetilde{Z''^2} = C_\phi^{-1} (\Delta_\alpha^{\text{TCR}})^2 \nabla \tilde{Z} \cdot \nabla \tilde{Z} \quad (8)$$

$$\chi = 2 \frac{\Gamma_\alpha}{\rho} \nabla \tilde{Z} \cdot \nabla \tilde{Z} \quad (9)$$

where Z is the mixture fraction; $\Delta_\alpha^{\text{TCR}}$ is the spatial mixing scale for scalar α at the current space-time location, corresponding to the perfectly stirred portion after grid partitioning; μ_{SGS} is obtained from the turbulence model; C_ϕ is determined via the dynamic approach³⁹:

$$C_\phi^{-1} = \frac{\langle LM \rangle}{\langle M^2 \rangle} \quad (10)$$

where L is the Leonard term and M is the model term, and is determined using test filters³⁹. This approach removes the need to specify C_ϕ and has been applied and validated in jet flames^{37,38}. Note that only mixture fraction Z or passive scalars are recommended for the test filter, as they primarily relate to turbulent transport and mixing without drastic spatiotemporal variations. The final mixing timescale is expressed as:

$$\tau_{\text{SGS}}^{\text{TCR}} = \frac{\widetilde{Z''^2}}{\chi} = \frac{\bar{\rho} (\Delta_\alpha^{\text{TCR}})^2}{C_\phi \Gamma_\alpha} \quad (11)$$

2.2.2. Multi-layer PaSR for mixing target determination

The critical implementation aspect of the TCR mixing model involves determining the equilibrium mixing target ψ_α^n . This is achieved through a modified PaSR framework that partitions the computational cell into two conceptual zones: a PSR zone with uniform composition (volume fraction κ) and an unmixed zone. Traditional PaSR models compute κ through turbulence and chemical timescale expressions, then estimate the filtered reaction rate as $\widetilde{\omega_\alpha(\phi)} = \kappa \omega_\alpha^{\text{PSR}} = \kappa \omega_\alpha(\tilde{\phi})$. However, this formulation exhibits theoretical limitations since $\kappa \leq 1$ inherently constrains $\widetilde{\omega_\alpha} \leq \omega_\alpha^{\text{PSR}}$ —a relationship valid only for cells containing purely reactive mixtures. The presence of inert diluents or reaction products can induce filtered reaction rates exceeding PSR predictions due to multi-scalar mixing effects unaccounted for in conventional PaSR approaches.

To address this deficiency, we develop a multi-layer PaSR framework that decouples dilution effects from turbulence-chemistry interactions through hierarchical treatment of scalar mixing processes. The enhanced formulation enables independent resolution of inert species transport and reactive scalar evolution while maintaining the turbulence-chemistry equilibrium targeting essential to TCR modeling.

Within this framework, consider a representative reaction $\alpha + \beta \rightarrow \gamma$ without catalytic effects for modeling simplification. Reactants α and β are grouped into mixture m , while product γ and non-reactive scalars Ω are grouped into mixture n . Let η_α denote the combined molar fraction of reactants α and β :

$$\eta_\alpha = X_\alpha + X_\beta \quad (12)$$

where X_α and X_β represent the mole fractions of species α and β , respectively. The quantity η_α effectively characterizes the total proportion of reactants for the current reaction within the control volume, satisfying $0 < \eta_\alpha \leq 1$.

Assuming equivalent mixing homogeneity between m - n and α - β pairs within the control volume, the mixing completeness is quantified by κ_α , the volume fraction of the PSR zone, as is shown in Fig. 3, $0 \leq \kappa_\alpha \leq 1$.

This parameterization enables construction of a multi-level mixing architecture for the target reaction, where κ_α governs both inter-mixture (m - n) and intra-reactant (α - β) homogenization processes, as is shown in Fig. 4.

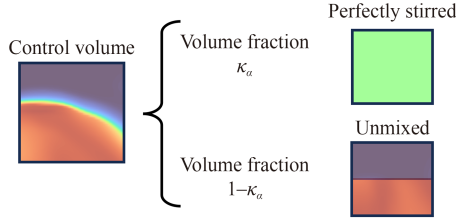


Fig. 3 Schematic diagram illustrating division into a perfectly stirred zone and an unmixed zone.

The control volume is initially partitioned into two zones based on m - n mixing homogeneity: Zone 1 with volume fraction κ_α containing uniformly mixed m - n , and Zone 2 with volume fraction $1 - \kappa_\alpha$ maintaining complete segregation between m and n . The total filtered reaction rate is expressed as:

$$\overline{\dot{\omega}_\alpha(\phi)} = \kappa_\alpha \dot{\omega}_1 + (1 - \kappa_\alpha) \dot{\omega}_2 \quad (13)$$

Within Zone 1, the reactive mixture undergoes secondary partitioning based on α - β mixing completeness. A fraction κ_α of Zone 1 constitutes Zone 1A where full α - β mixing enables maximum reaction rate $\dot{\omega}_{1A} = \dot{\omega}(\tilde{\phi})$, while the remaining $1 - \kappa_\alpha$ forms Zone 1B—a non-reactive zone with zero chemical activity. This yields:

$$\dot{\omega}_1 = \kappa_\alpha \dot{\omega}_{1A} + (1 - \kappa_\alpha) \dot{\omega}_{1B} = \kappa_\alpha \dot{\omega}_\alpha(\tilde{\phi}) \quad (14)$$

Zone 2 is subsequently divided proportionally to the reactant volume fraction η_α , creating Zone 3 (fraction η_α) containing pure reactants and Zone 4 (fraction $1 - \eta_\alpha$) containing mixture n . The reaction rate in Zone 2 becomes:

$$\dot{\omega}_2 = \eta_\alpha \dot{\omega}_3 + (1 - \eta_\alpha) \dot{\omega}_4 = \eta_\alpha \dot{\omega}_3 \quad (15)$$

Tertiary partitioning of Zone 3 introduces Zone 3A (fraction κ_α) with perfectly mixed α - β reactants and Zone 3B (fraction $1 - \kappa_\alpha$) exhibiting reactant segregation. The corresponding rate expression is:

$$\dot{\omega}_3 = \kappa_\alpha \dot{\omega}_{3A} + (1 - \kappa_\alpha) \dot{\omega}_{3B} = \eta_\alpha \dot{\omega}_{3A} \quad (16)$$

Assuming equivalent enthalpy distributions between diluted (Zone 1A) and undiluted (Zone 3A) reactant mixtures, the Arrhenius rate relationship establishes:

$$\frac{\dot{\omega}_{3A}}{\dot{\omega}(\tilde{\phi})} = \frac{Y_{\alpha,3A} Y_{\beta,3A}}{Y_{\alpha,1A} Y_{\beta,1A}} = \frac{1}{\eta_\alpha^2} \Rightarrow \dot{\omega}_{3A} = \frac{1}{\eta_\alpha^2} \dot{\omega}_\alpha(\tilde{\phi}) \quad (17)$$

Combining these hierarchical partitions yields the final filtered reaction rate:

$$\overline{\dot{\omega}_\alpha(\phi)} = \left[\kappa_\alpha^2 + \frac{\kappa_\alpha(1 - \kappa_\alpha)}{\eta_\alpha} \right] \dot{\omega}_\alpha(\tilde{\phi}) \quad (18)$$

When chemical reactions occur ($\dot{\omega}_\alpha(\tilde{\phi}) \neq 0$; otherwise $\kappa_\alpha = 1$ by definition), Eq. (18) can be reformulated to establish the relationship between the degree of mixing homogeneity κ_α (quantified as the PSR volume fraction in PaSR), the filtered-to-PSR reaction rate ratio $\frac{\overline{\dot{\omega}_\alpha(\phi)}}{\dot{\omega}_\alpha(\tilde{\phi})}$, and the reactants volume fraction η_α :

$$\kappa_\alpha = \begin{cases} \frac{1 \pm \sqrt{1 - 4\eta_\alpha(1 - \eta_\alpha) \frac{\overline{\dot{\omega}_\alpha(\phi)}}{\dot{\omega}_\alpha(\tilde{\phi})}}}{2(1 - \eta_\alpha)}, & 0 < \eta_\alpha < 1 \\ \frac{\overline{\dot{\omega}_\alpha(\phi)}}{\dot{\omega}_\alpha(\tilde{\phi})}, & \eta_\alpha = 1 \end{cases} \quad (19)$$

Eq. (19) demonstrates critical limiting behavior when analyzed across the reactant volume fraction (η_α) spectrum. In the absence of diluent effects ($\eta_\alpha = 1$), the expression for κ_α collapses to the classical PaSR, where SGS scalar inhomogeneity linearly reduces $\frac{\overline{\dot{\omega}_\alpha(\phi)}}{\dot{\omega}_\alpha(\tilde{\phi})}$ below the fully mixed state value.

Introduction of diluents ($\eta_\alpha < 1$) fundamentally alters this behavior, enabling both enhancement and suppression of $\frac{\overline{\dot{\omega}_\alpha(\phi)}}{\dot{\omega}_\alpha(\tilde{\phi})}$ relative to homogeneous conditions depending on the reactant concentrn. The dependence of $\frac{\overline{\dot{\omega}_\alpha(\phi)}}{\dot{\omega}_\alpha(\tilde{\phi})}$ on mixing homogeneity exhibits distinct regimes governed by the reactant volume fraction η_α , as shown in Fig. 5. For $0 < \eta_\alpha < 0.5$, SGS scalar inhomogeneity may either increase or decrease the filtered reaction rate compared to the fully mixed state. When $0.5 \leq \eta_\alpha \leq 1.0$, improved mixing homogeneity monotonically enhances the filtered reaction rate, achieving maximum values at the perfectly mixed PSR condition ($\kappa_\alpha = 1$).

The quadratic formulation of Eq. (19) originates from the multi-layer mixing framework, where the non-negativity constraint of the discriminant $1 - 4\eta_\alpha(1 - \eta_\alpha) \frac{\overline{\dot{\omega}_\alpha(\phi)}}{\dot{\omega}_\alpha(\tilde{\phi})} \geq 0$ ensures physically realizable solutions. This condition inherently bounds the filtered-to-PSR reaction rate ratio by:

$$0 \leq \frac{\overline{\dot{\omega}_\alpha(\phi)}}{\dot{\omega}_\alpha(\tilde{\phi})} \leq \frac{1}{4\eta_\alpha(1 - \eta_\alpha)} \quad (20)$$

The lower bound preserves reaction directionality by preventing sign reversal of chemical production rates, while the upper limit reflects the maximum attainable reaction rate under optimal mixing conditions dictated by reactants volume fraction η_α . Physically, this implies that SGS scalar mixing can neither reverse chemical transformation directions nor exceed reaction rates achievable in the ideally mixed state determined by local reactant availability.

The parameter κ_α exhibits an approximately inverse correlation with the Da number, decreasing as chemical reactions increasingly dominate over turbulent mixing. This behavior aligns with traditional PaSR models where κ_α is determined by timescale ratios. When solving Eq. (19), the Da number corresponding to peak filtered-to-PSR reaction rate ratios is first identified. The sign selection in the solution is then determined by comparing local Da values to this characteristic peak condition.

In Eq. (19), the reactant volume fraction η_α can be directly obtained from the filtered scalar field. The PSR reaction rate $\dot{\omega}_\alpha(\tilde{\phi})$ is computed by inputting the filtered scalar concentrns into the chemical reaction solver, which also provides initial reaction rate estimates for each stochastic field in the TPDF approach. The filtered reaction rate $\overline{\dot{\omega}_\alpha(\phi)}$ is derived from TPDF solutions, enabling determination of κ_α for reactants of a specific reaction at a given spatiotemporal location. When multiple reactions share α as a reactant, the mean value of κ_α obtained by substituting their respective reaction rates into Eq. (19) serves as the mixing homogeneity indicator for scalar α . This indicator subsequently determines the micro-mixing target ψ_α^n of the stochastic variable ξ_α^n in Eq. (7).

2.2.3. Expression of TCR model

The TCR model incorporates two assumptions: First, it extends the partitioning hypothesis of the PaSR conceptual framework, i.e., the control volume is abstracted into two constituent parts: one being a PSR, and the other having completely unmixed scalars. Second, molecular diffusion and chemical reactions attain dynamic balance within the control volume, i.e., the close-parallel assumption⁷⁰, this assumption forms the foundation for quantifying chemical inhomogeneities to establish the mixing model.

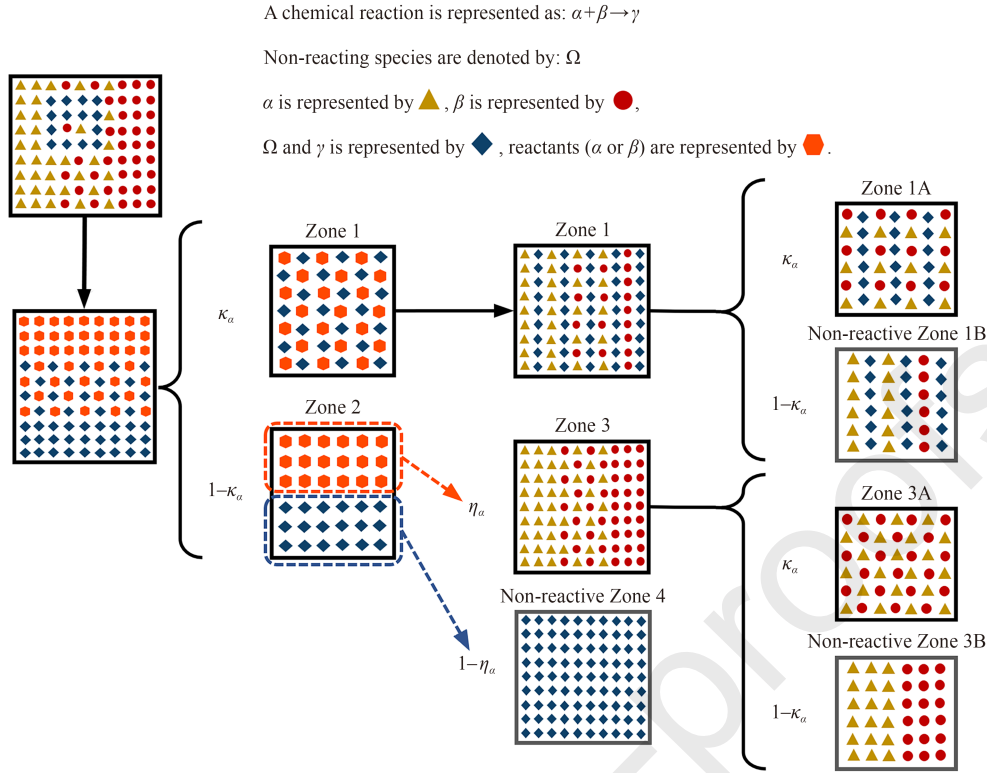


Fig. 4 Schematic of multi-layer PaSR framework architecture.

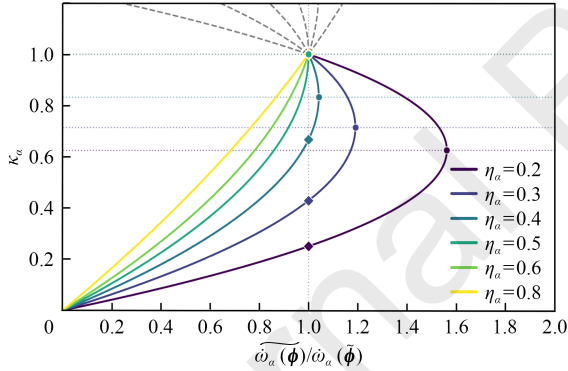


Fig. 5 Mixing homogeneity κ_α versus filtered-to-PSR reaction rate ratio $\frac{\omega_\alpha(\Phi)}{\omega_\alpha(\bar{\Phi})}$ at different reactant volume fractions η_α .

Following the PaSR conceptual framework, the control volume is partitioned into two zones: a perfectly stirred zone occupying volume fraction κ_α and a non-mixed zone with fraction $1 - \kappa_\alpha$, both maintaining identical volumetric mean concentrations of scalar α , as shown in Fig. 3. Chemical reactions and scalar variance within the control volume are determined by these zones. Micro-mixing in the control volume can be treated as a combination of micro-mixing processes occurring in both zones.

Regarding variance evolution of stochastic variable ξ_α^n : In the ESF-TPDF method, the Wiener term generates stochastic transport in physical space, causing ξ_α^n to deviate from its mean value within the grid, thereby increasing variance. The direction and magnitude of deviation depend on the stochastic field and mean scalar values in the current and adjacent grids. Notably, this variance increase by the Wiener term requires no SGS scalar distribution information. Conversely, the micro-mixing term reduces variance through scalar-

space mixing. The TCR model combines these opposing effects to produce stochastic variables that correctly represent scalar variance in the control volume, ensures the sample space of ξ_α^n matches the actual combustion field, as shown in Fig. 2.

In the perfectly stirred zone, variance approaches zero, requiring full micro-mixing action to drive variables toward the mean (consistent with IEM model behavior). In the flamelet-like unmixed zone, variance remains maximized with negligible micro-mixing. Combining these zones by volume fraction yields the micro-mixing target ψ_α^n . For the n -th stochastic field, ξ_α^n evolves to attain the filtered mean $\bar{\phi}_\alpha$ in the stirred zone while retaining its initial value in the non-mixed zone. The micro-mixing target ψ_α^n is formulated as:

$$\psi_\alpha^n = \kappa_\alpha \bar{\phi}_\alpha + (1 - \kappa_\alpha) \xi_\alpha^n \quad (21)$$

Substituting Eq. (21) into Eq. (7) yields the micro-mixing term expression for this TCR model within the ESF-TPDF methodology:

$$\mathcal{M} = -\frac{\bar{\rho}}{2\tau_{\text{SGS}}^{\text{TCR}}} (\xi_\alpha^n - \psi_\alpha^n) dt = -\frac{\bar{\rho}\kappa_\alpha}{2\tau_{\text{SGS}}^{\text{TCR}}} (\xi_\alpha^n - \bar{\phi}_\alpha) dt \quad (22)$$

Note that as $dt \rightarrow 0$, Eq. (22) describes the transient change of the stochastic variable ξ_α^n . Although the relaxation direction of ξ_α^n aligns with the mean, the relaxation target in the TCR model is ψ_α^n rather than $\bar{\phi}_\alpha$. Additionally, since micro-mixing in the TCR model occurs in the PSR zone with volume fraction κ_α , the spatial scale of micro-mixing can be derived from the filter scale Δ as $\Delta_\alpha^{\text{TCR}} = \sqrt[3]{\kappa_\alpha} \Delta$. Thus Eq. (11) can be expressed as:

$$\tau_{\text{SGS}}^{\text{TCR}} = \frac{\bar{\rho}\kappa_\alpha^{\frac{2}{3}}\Delta^2}{C_\phi\Gamma_\alpha} \quad (23)$$

Although a decrease in κ_α reduces the spatial scale of micro-mixing, which in turn decreases the mixing timescale (accelerating mixing), the expression shows that the effect of κ_α on the mixing timescale is weaker than its effect on the mixing mode Eq. (22). Thus, overall, a smaller κ_α results in a lower rate of change of ξ_α^n , corresponding to higher variance after micro-mixing.

The parameter κ_α in the TCR model is recursively determined from the filtered-to-PSR reaction rate ratio obtained in preceding computational steps using Eq. (19) and subsequently applied to the current TPDF calculation cycle. This treatment eliminates empirical model parameters while enabling self-recursive closure as demonstrated by the mechanism illustrated in Fig. 6.

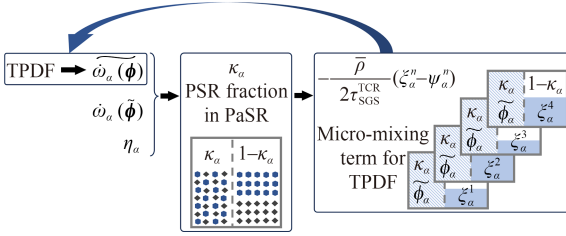


Fig. 6 Schematic of TCR model implementation in ESF-TPDF framework with 4 stochastic fields.

2.2.4. Discussion of performance criteria

Discussion of TCR model performance criteria comprises two aspects. First, the parameter κ_α governing control volume partitioning in TCR is compared with its counterpart in PaSR methods. Second, characteristic differences between TCR and IEM are examined.

Firstly, regarding κ_α modeling, conventional PaSR approaches derive κ_α from expressions combining chemical reaction and turbulent mixing timescales. This methodology presents two significant limitations. First, definitions of chemical reaction and turbulent mixing timescales lack precision. Second, chemical reaction timescales typically quantify reaction rates but cannot characterize SGS reaction uniformity.

Notably, the modulation factor $0 \leq \kappa_\alpha \leq 1$ does not imply uniform chemical inhibition of mixing in the TCR model. As Eq. (19) demonstrates, micro-mixing rates remain identical to non-reacting cases when filtered reaction rates $\bar{\omega}_\alpha(\phi)$ match PSR references $\omega_\alpha(\phi)$, regardless of chemical intensity. Thus κ_α depends not on reaction strength but on SGS inhomogeneity in reaction rates, capturing how local flame structure heterogeneity modulates mixing rather than chemical kinetics. This fundamentally differs from conventional PaSR approaches where $\kappa_{\text{Conv}} = \tau_{\text{Chem}} / (\tau_{\text{Chem}} + \tau_{\text{Turb}})$, as shown in Fig. 7.

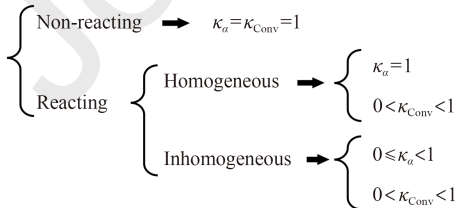


Fig. 7 Differences between conventional PaSR and proposed multi-layer PaSR in reacting and non-reacting zones.

E.g., for premixed combustion with uniform temperature, pressure, density and concentrations, reaction rates exhibit homogeneous SGS distributions. Conventional PaSR models would incorrectly yield $\kappa_\alpha < 1$ in Eq. (22), implying artificial chemical

inhibition of mixing. In contrast, the proposed Eq. (19) correctly yields $\kappa_\alpha = 1$, maintaining mixing rates identical to non-reacting cases. Crucially, the TCR approach decouples direct chemical effects on mixing; instead, SGS reaction rate heterogeneity modulates mixing by indicating the existence and fractional extent of reaction-diffusion balanced gradient regions within control volumes.

Secondly, regarding characteristic differences between TCR and IEM, the primary cause of distinction stems from different relaxation targets for stochastic variables. For clarity, from the Lagrangian stochastic method perspective: stochastic variables are carried by stochastic particles in physical space. These particles may reside at arbitrary positions within the control volume, with those located in the PSR zone participating in micro-mixing while particles in the unmixed zone remain non-participatory. Assuming sufficient stochastic particles ($N \rightarrow \infty$), the rate-of-change statistics for scalars associated with these particles can be interpreted in scalar space as all stochastic variables undergoing micro-mixing, the mixing target comprises a combination of the mean value and initial stochastic variable values, as shown in Fig. 8.

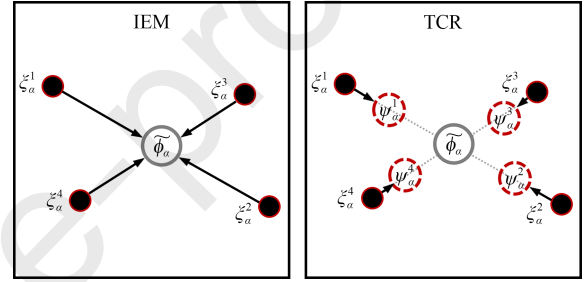


Fig. 8 Schematic diagram illustrating differences between TCR and IEM models from the perspective of Lagrangian stochastic method

The TCR model preserves the essential mathematical properties of IEM, maintaining the fundamental linear relaxation mechanism while introducing localized frequency modulation via κ_α . This retention ensures TCR inherits IEM's desirable characteristics: mass conservation, variance decay, boundedness, linearity, and independence - all essential for effective micro-mixing modeling.

The primary distinction between TCR and IEM lies in TCR's incorporation of local chemical effects on micro-mixing. Specifically, the spatially heterogeneous scalar distribution sustained by combined chemical-turbulent interactions influences variance in the sample space of stochastic variable ξ_α^n . Scalar distribution inhomogeneity within the control volume creates disparity between filtered reaction rate $\bar{\omega}_\alpha(\phi)$ and PSR reaction rate $\omega_\alpha(\phi)$, quantified as κ_α . Greater scalar inhomogeneity yields smaller κ_α , weakening TCR's variance reduction effect. This results in larger variance in ξ_α^n 's sample space, better matching physical reality, whereas IEM fails to correctly represent SGS scalar inhomogeneity.

A secondary distinction concerns TCR's micro-mixing target ψ_α^n being related to the local value ξ_α^n . This relationship arises because stochastic variables preferentially mix with neighboring scalar values, which typically approximate ξ_α^n . Crucially, ψ_α^n in Eq. (21) is explicitly defined, while the micro-mixing term in Eq. (7) permits implicit solution. This implementation links micro-mixing to ξ_α^n 's local value, conferring localness characteristics.

The tertiary distinction involves stochastic variability introduced through TCR's dependence on filtered reaction rates from TPDF solutions. This randomness originates from Wiener-process-driven scalar distributions in individual stochastic fields, propagating into the filtered-to-PSR reaction rate ratio and consequently inducing stochastic fluctuations in κ_α . In non-reacting regions

($\kappa_\alpha \rightarrow 1$), Gaussian relaxation remains unattainable due to the pre-served linear operator form, though this limitation doesn't impede performance in inhomogeneous flows¹⁸.

A comparative analysis of performance criteria between the TCR model and conventional mixing models is presented in Table 1, following the evaluation framework established by Subramaniam and Pope¹⁸. Model compatibility with the ESF-TPDF framework is assessed by identifying particle-information requirements.

Table 1 Mixing model assessment of TCR, IEM, MC and EMST⁷¹.

Feature	TCR	IEM	MC	EMST
Conservation of means	●	●	●	●
Decay of variance	●	●	●	●
Boundedness	●	●	●	●
Linearity and independence	●	●	●	○
Differential diffusion	●	●	●	○
Length scales dependence	●	●	●	○
Localness	●	●	●	●
Relaxation to Gaussian	○	○	●	○
Lagrangian-independent	●	●	○	○
Computational complexity	$O(N_{sc})$	$O(N_{sc})$	$O(N_{sc})$	$O(N_{sc}^2)$

Notes: Symbols ●, ○, and ○ refer to requirements that are completely, partly and not fulfilled respectively.

3. Configurations and settings

3.1. Vitiated coflow burner

Previous studies^{28,38,72} have demonstrated the necessity of parameter adjustments in conventional mixing models like IEM for accurate flame lift-off height predictions when simulating turbulent lifted flames using TPDF methods. This requirement particularly applies to the SGS mixing frequency parameter C_ϕ , indicating that lifted flame simulations effectively expose differences between micro-mixing models in TPDF approaches. The present study employs Cabra's Vitiated Coflow Burner (VCB) with hydrogen fuel diluted by nitrogen⁷³, conducting three-dimensional unsteady numerical simulations within the LES and ESF-TPDF frameworks. Comparative analyses are performed between the IEM model ($C_\phi = 2.0$, following common recommendations^{63,74–76}) and the proposed TCR model, with results validated against experimental data. The VCB geometry is shown in Fig. 9.

The VCB configuration features a central fuel nozzle with the diameter $D = 4.57$ mm extending 70 mm above a coaxial perforated plate, delivering H_2/N_2 fuel mixture with 1 : 3 volumetric ratio at 305 K with 107 m/s bulk velocity. The coflow consists of hot combustion products from a lean H_2 /air premixed flame ($\phi \approx 0.367$) maintained at 1045 K with 3.5 m/s bulk velocity, where an exit collar prevents ambient air entrainment. Experimental measurements employed Raman and Rayleigh scattering techniques for temperature and mass fractions of N_2 , O_2 , H_2O , and H_2 , with OH mass fractions obtained via laser-induced fluorescence. Measurement uncertainties are 1% for temperature, 5% for H_2O , and 12% for OH⁷³. Key operational parameters include Reynolds number $Re = 23600$, stoichiometric mixture fraction $Z_{st} = 0.474$ (Bilger formulation⁷⁷), and measured lift-off height is $H_{lift} = 10D$. Detailed inlet specifications are provided in Table 2.

3.2. TPDF simulation settings

The ESF method for TPDF simulations employs an Eulerian description aligned with finite-volume fluid solvers, eliminating the

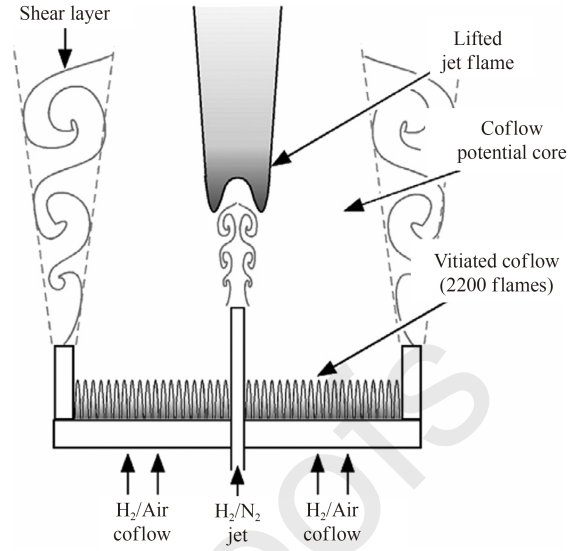


Fig. 9 Combustor geometry of Cabra et al.⁷³ for lifted flame.

Table 2 Detailed inlet specifications of VCB⁷³.

Parameter	Jet	Coflow
Velocity (m/s)	107.0	3.5
Temperature (K)	305	1045
H_2 mole fraction	0.2537	0.0003
O_2 mole fraction	0.0021	0.1474
H_2O mole fraction	0.0015	0.7534
N_2 mole fraction	0.7427	0.0989

need for Lagrangian particle tracking and thereby reducing algorithmic complexity for practical engineering applications. However, this particle-free framework precludes implementation of mixing models requiring Lagrangian particle information, such as MC and EMST. To evaluate the proposed TCR mixing model's performance, the present study implements both TCR and IEM models within the ESF-TPDF framework for VCB simulations, with results rigorously validated against experimental measurements.

The LES framework employs the dynamic Smagorinsky model⁷⁸ to determine SGS stresses, and the computational time step is dynamically adjusted by constraining the Courant-Friedrichs-Lewy (CFL) number within the range of 0.25 to 0.35. Chemical reaction rates for both PSR references and stochastic fields in the TPDF method are solved using a variable-coefficient Ordinary Differential Equation (ODE) solver⁷⁹. All simulations incorporate a validated hydrogen oxidation mechanism⁸⁰ and account for differential diffusion effects through dynamic Schmidt number (σ_α) determination⁵⁴.

The computational domain comprises a cylindrical volume extending $44D$ upstream from the fuel jet exit with a radius of $10.5D$. Grid independence studies⁵⁴ were conducted using three mesh resolutions: 1.62 million, 2.16 million, and 2.70 million cells. Cold-flow simulations assessed grid adequacy through Pope's LES resolution criterion⁸¹, which requires the ratio of SGS turbulent kinetic energy (k_{SGS}) to total turbulent kinetic energy (k_{total}) to remain below 0.2 throughout the domain. Both the 2.16 million and 2.70 million cell configurations satisfied this criterion globally, with negligible differences in velocity and mixture fraction predictions. Consequently, the 2.16 million cell mesh was selected for combustion simulations, providing optimal resolution while maintaining

computational efficiency for the parametric studies.

Simulations utilized 120 CPU cores in parallel, maintaining an average time step of $1.11 \mu\text{s}$. After 30000 time steps (approximately 25.64 flow-through times), the combustion field is considered to reach statistical stationarity. Data collection subsequently spanned 120000 steps (approximately 102.57 flow-through times) to obtain converged statistics. Computational cost comparisons within the ESF-TPDF framework (4 stochastic fields) revealed the IEM model required 4.01 hours per 1000 steps, while the proposed TCR model averaged 4.49 hours per thousand steps, indicating a 11.97% increase primarily attributed to additional PSR reaction rate calculations. This overhead is partially mitigated by utilizing PSR rates as initial estimates for stochastic field reaction rates. Notably, the relative computational penalty decreases with larger stochastic field ensembles, as PSR calculation costs become amortized across more fields.

4. Results and discussion

4.1. Flame characteristics and species profiles

Fig. 10 presents transient temperature fields obtained using both IEM and TCR models coupled with the ESF-TPDF method. Both simulations exhibit distinct flame lift-off in the VCB configuration, manifesting clear separation between high-temperature regions and the nozzle exit.

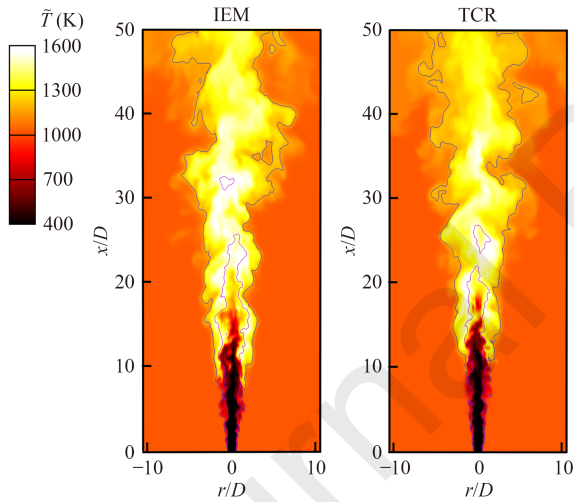


Fig. 10 Instantaneous resolved temperature comparison between IEM (left) and TCR (right) mixing model (the inner isoline corresponds to the stoichiometric mixture fraction $Z_{\text{st}} = 0.474$ and the outer isoline shows the most reactive mixture fraction $Z_{\text{mr}} = 0.054$).

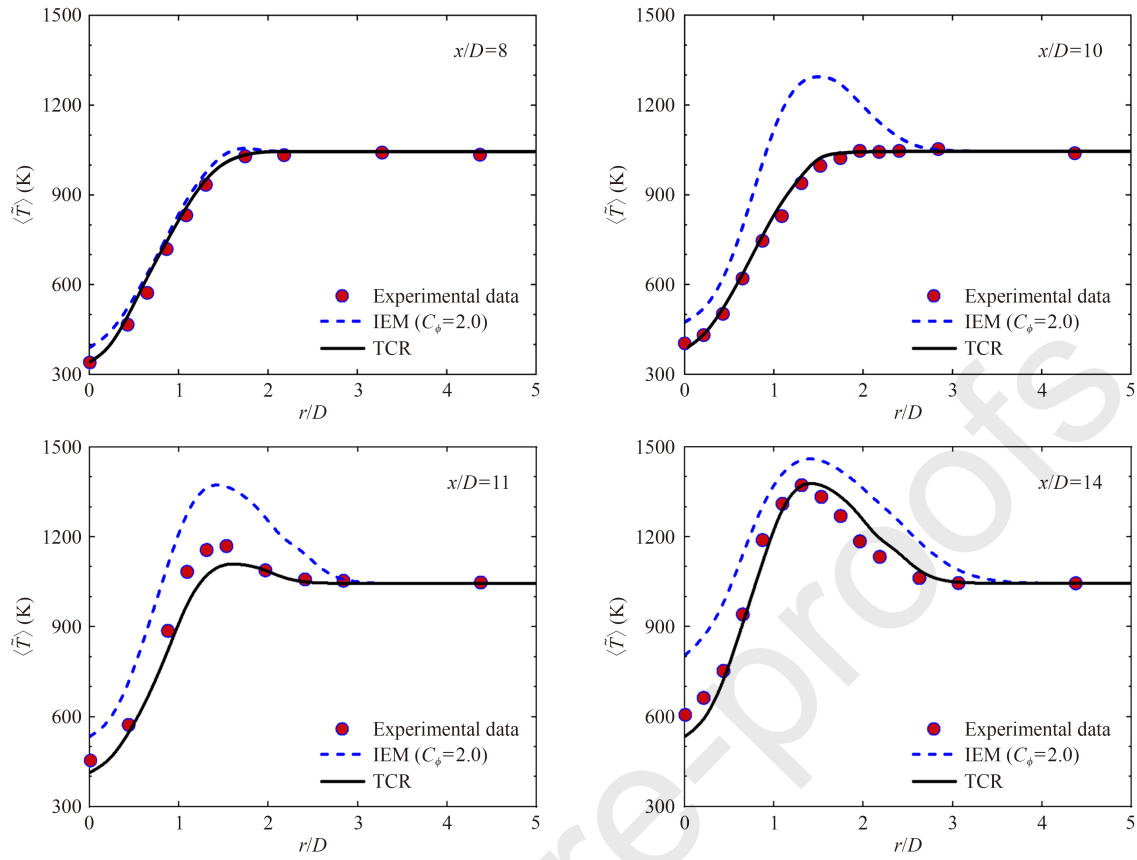


Fig. 11 Model predictions versus experimental measurements⁷³ of radial mean temperature profiles at multiple axial locations

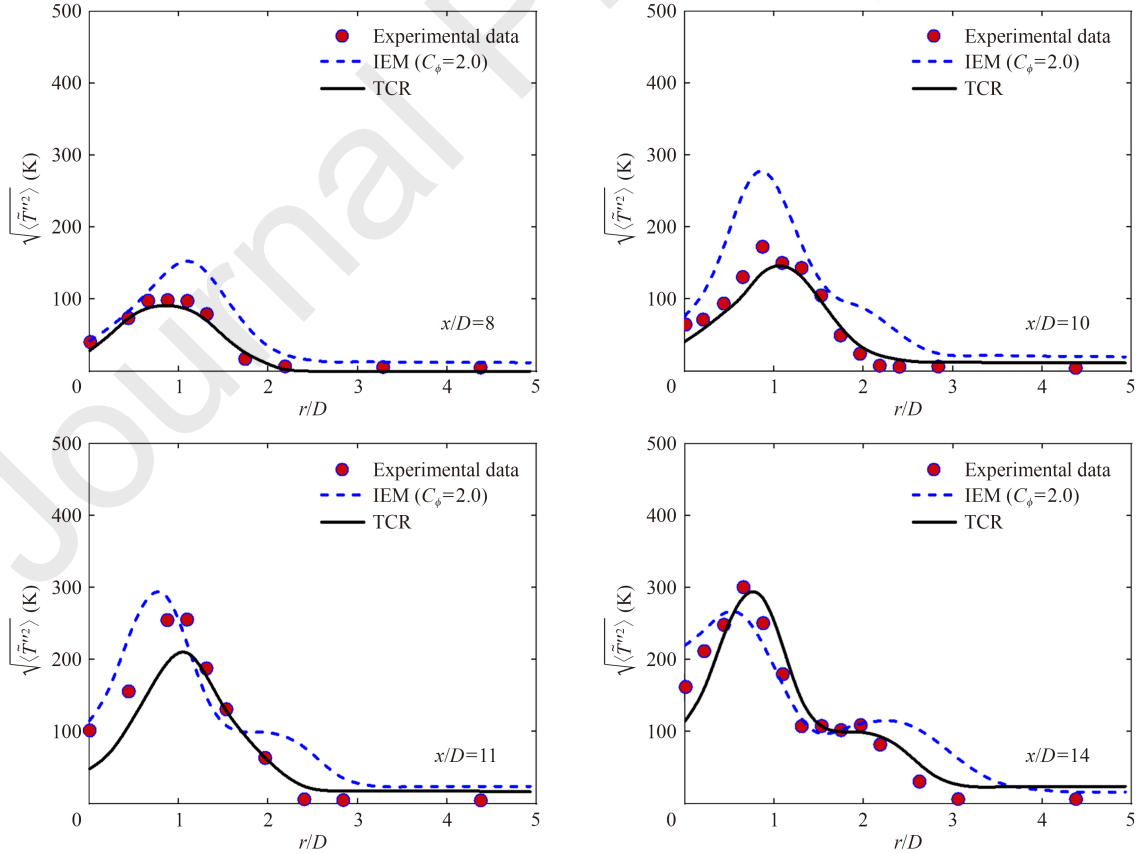


Fig. 12 Model predictions versus experimental measurements⁷³ of radial temperature RMS fluctuations at multiple axial locations

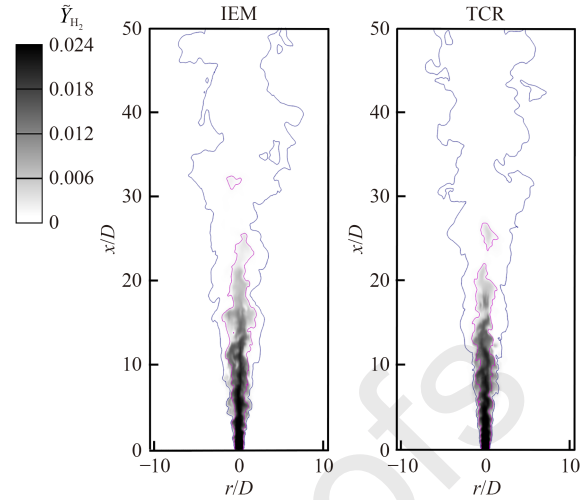
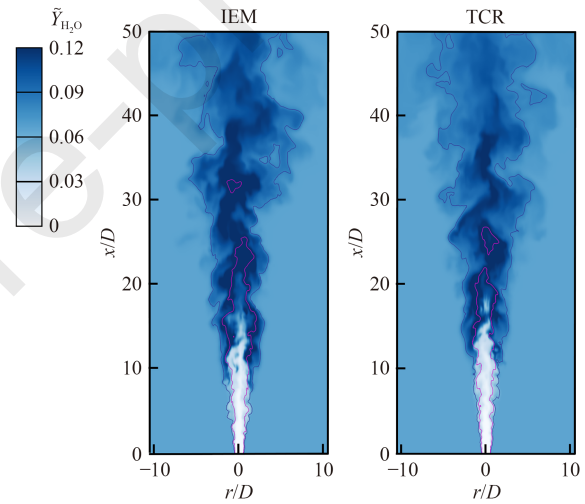
Table 3 Radial-averaged relative error of mean temperature.

Axial location	Radial-averaged relative error(%)	
	IEM	TCR
$x/D = 8$	3.28	1.99
$x/D = 10$	16.02	1.44
$x/D = 11$	11.43	3.34
$x/D = 14$	10.08	3.08
Average value	10.20	2.46

Table 4 Radial-averaged relative error of temperature RMS fluctuations.

Axial location	Radial-averaged relative error(%)	
	IEM	TCR
$x/D = 8$	26.66	7.02
$x/D = 10$	31.49	7.70
$x/D = 11$	15.41	9.89
$x/D = 14$	9.05	7.71
Average value	20.65	8.08

The predicted lift-off heights approximate $10D$, consistent with experimental measurements. Although experimental lift-off height measurements may exhibit uncertainty⁷² due to their well-documented sensitivity to coflow temperature, this pronounced environmental dependence establishes lifted flames as particularly effective test cases for evaluating micro-mixing models, as it accentuates inter-

**Fig. 13** Instantaneous resolved H_2 mass fraction comparison between IEM (left) and TCR (right) mixing model (the inner isoline corresponds to the stoichiometric mixture fraction $Z_{st} = 0.474$ and the outer isoline shows the most reactive mixture fraction $Z_{mr} = 0.054$).**Fig. 14** Instantaneous resolved H_2O mass fraction comparison between IEM (left) and TCR (right) mixing model (the inner isoline corresponds to the stoichiometric mixture fraction $Z_{st} = 0.474$ and the outer isoline shows the most reactive mixture fraction $Z_{mr} = 0.054$).

model differences. The marginally higher lift-off height predicted by TCR relative to IEM indicates that the micro-mixing treatment in the IEM model promotes combustion reactions and enhances flame stabilization more effectively.

Fig. 11 compares radial distributions of time averaged temperature at four critical axial sections near the flame base ($x/D = 8, 10, 11, 14$). At $x/D = 8$, neither simulations nor experimental data exhibit significant combustion induced temperature rise. At $x/D = 10$, results obtained with the TCR mixing model remain consistent with experimental measurements in showing negligible temperature increase, whereas the IEM model predicts a distinct high temperature region exceeding 1300 K near $r/D = 1.5$, indicating premature combustion initiation below $x/D = 10$. At the two higher sections ($x/D = 11, 14$), temperatures predicted by IEM substantially exceed both TCR predictions and experimental values. This demonstrates that IEM's mixing treatment artificially enhances combustibility, initiating reactions further upstream and

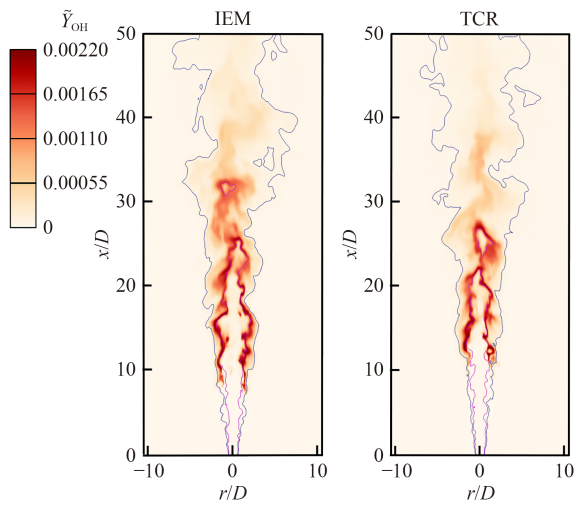


Fig. 15 Instantaneous resolved OH mass fraction comparison between IEM (left) and TCR (right) mixing model (the inner isoline corresponds to the stoichiometric mixture fraction $Z_{st} = 0.474$ and the outer isoline shows the most reactive mixture fraction $Z_{mr} = 0.054$).

consequently introducing greater temperature prediction errors. As quantified in Table 3, the mean temperature prediction error under the ESF-TPDF framework is 10.20% for IEM versus 2.46% for the proposed TCR model.

Fig. 12 compares radial distributions of temperature Root Mean Square (RMS) fluctuations at key axial sections. Near $x/D = 8$ and $r/D = 1$, the observed RMS peak primarily originates from mixing between the central jet and hot coflow. At $x/D = 10, 11$, combustion reactions amplify thermal gradients, resulting in elevated RMS fluctuations compared to non-reacting conditions. The bimodal RMS distribution at $x/D = 14$ characterizes the thickness of the reaction-sustained high-temperature zone, with fluctuation peaks occurring at high-gradient interfaces between burned and unburned regions. IEM's erroneous prediction of premature high-temperature regions at $x/D = 10$ and systematic overestimation of temperatures at $x/D = 11, 14$ (Fig. 11) consequently produce exaggerated RMS fluctuation levels. Quantitative error analysis in Table 4 confirms this discrepancy, showing mean RMS prediction errors of 20.65% for IEM versus 8.08% for the proposed TCR model.

Figs. 13-15 present instantaneous resolved concentrations of hydrogen fuel, water vapor, and hydroxyl radical, respectively, while Figs. 16-18 comparatively display radial distributions of their mean mass fractions simulated using IEM and TCR models against experimental measurements. The hydrogen mass fraction distributions, characterizing fuel jet development and dissipation, show minimal divergence between IEM and TCR simulations, though Table 5 indicates marginally higher prediction errors for IEM.

The H_2O concentration profiles follow similar distribution patterns to temperature, with corresponding error characteristics mirroring temperature prediction trends. Hydroxyl radical concentrations, serving as markers for intense reaction zones, exhibit heightened sensitivity to both turbulent combustion modeling and chemical kinetic mechanisms. As a low-concentration intermediate species, OH presents greater prediction challenges than major reactants or products. When employing identical chemical mechanisms, IEM produces substantially broader high-OH concentration regions compared to TCR. Table 5 reveals particularly severe OH prediction inaccuracies for IEM, exceeding 300% error at $x/D = 8$ and $x/D = 10$ sections, culminating in an average error of 179.57%. The proposed TCR model achieves significant improvement, reducing

the average OH prediction error to 16.81%.

Complementary RMS fluctuation analyses in Figs. 19-21 further elucidate model performance disparities. As is shown in Table 6, for H_2 , both models exhibit comparable RMS prediction accuracy with errors generally below 10%, consistent with their similar mean concentration predictions. The H_2O RMS profiles demonstrate TCR's superior capability in capturing product formation dynamics. In addition, TCR corrects excessive OH fluctuations of IEM, by nearly an order of magnitude upstream at $x/D = 8$, and with error reductions exceeding 75% at $x/D = 10$ and $x/D = 11$. The residual OH RMS discrepancy at $x/D = 14$ suggests localized modeling challenges in downstream flame regions, yet TCR maintains substantially lower domain-averaged RMS errors across all species.

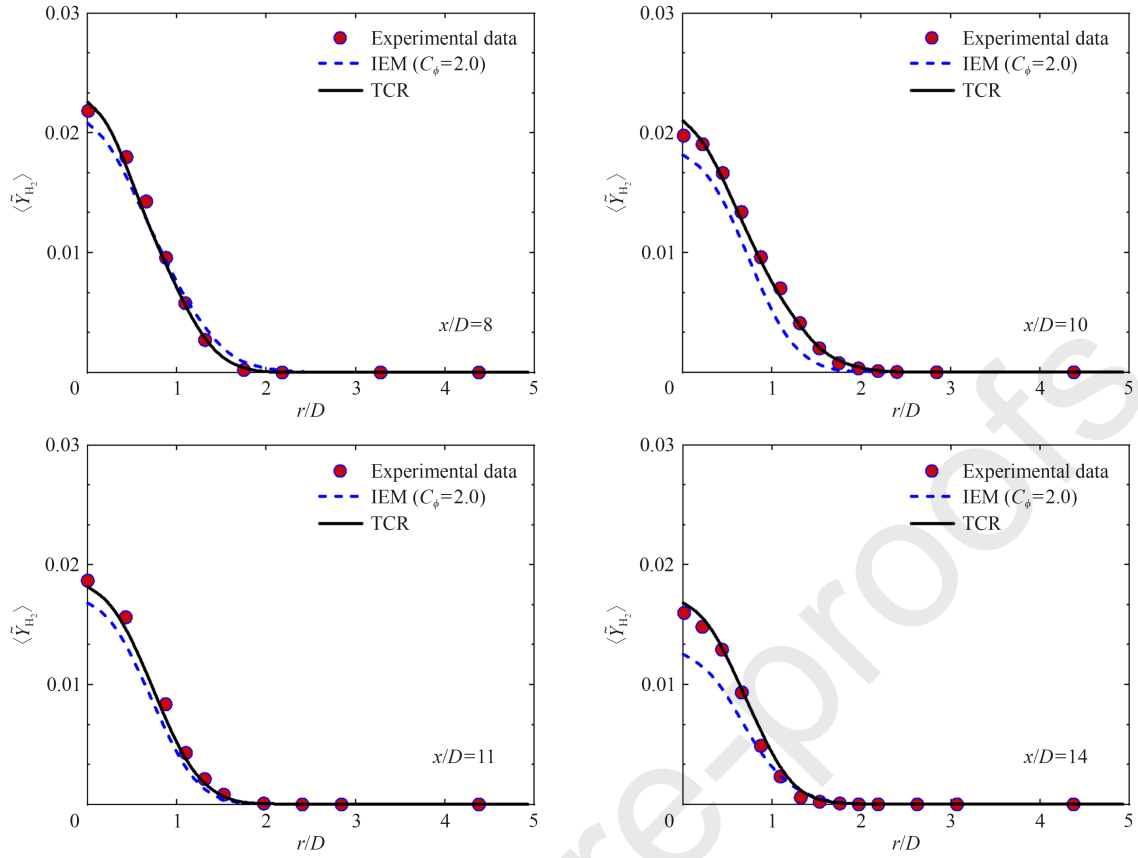


Fig. 16 Model predictions versus experimental measurements⁷³ of H_2 mass fraction profiles at multiple axial locations

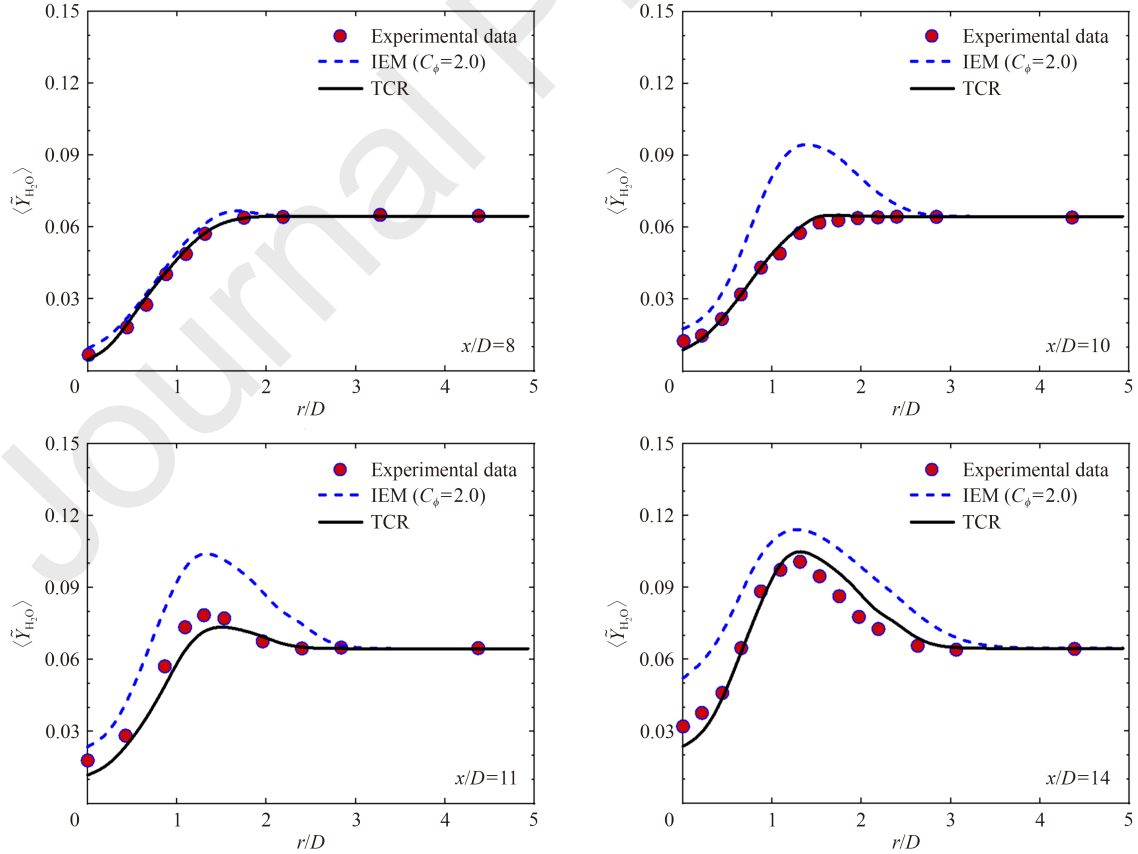


Fig. 17 Model predictions versus experimental measurements⁷³ of H_2O mass fraction profiles at multiple axial locations

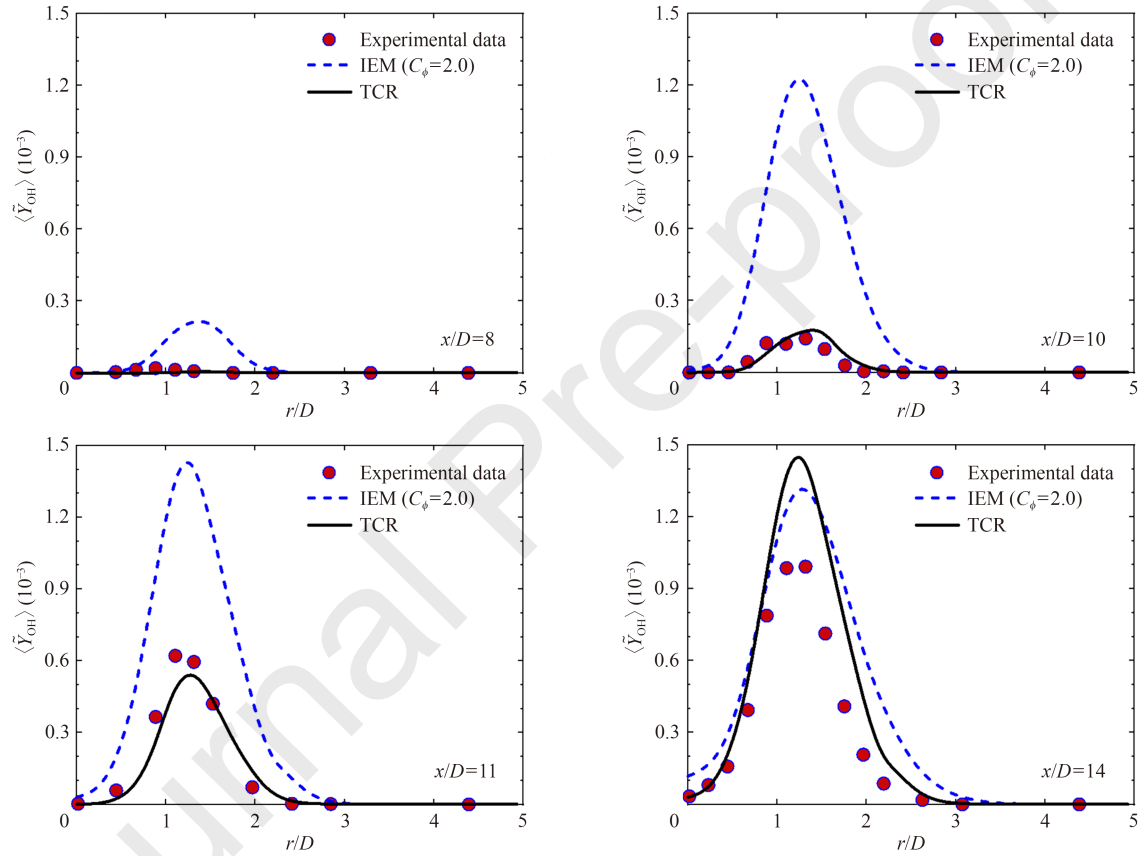


Fig. 18 Model predictions versus experimental measurements⁷³ of OH mass fraction profiles at multiple axial locations

Table 5 Radial-averaged relative error of species mean mass fractions.

Species	Axial location	Radial-averaged relative error(%)	
		IEM	TCR
H ₂	$x/D = 8$	3.09	1.86
H ₂	$x/D = 10$	6.79	1.62
H ₂	$x/D = 11$	6.37	3.03
H ₂	$x/D = 14$	9.51	2.21
H ₂ O	$x/D = 8$	4.35	1.72
H ₂ O	$x/D = 10$	27.45	2.18
H ₂ O	$x/D = 11$	20.74	4.88
H ₂ O	$x/D = 14$	16.05	5.06
OH	$x/D = 8$	330.18	31.51
OH	$x/D = 10$	308.73	14.02
OH	$x/D = 11$	57.35	4.74
OH	$x/D = 14$	22.02	16.97

Distinct from other locations, RMS fluctuation errors at $x/D = 14$ exhibit comparable magnitudes between IEM and TCR models. This convergence likely occurs because combustion at elevated positions attains a more intense and stabilized state, where micro-mixing effects become secondary to dominant factors: chemical reaction kinetics and boundary condition specifications. Consequently, simulation accuracy in flame root regions—characterized by reaction initiation and instability—demonstrates higher sensitivity to micro-mixing formulations. Improving predictive capability at elevated heights necessitates refined chemical kinetic mechanisms and boundary condition implementations rather than micro-mixing model optimization.

4.2. Conditional statistics

Scatter plots of temperature versus mixture fraction at axial locations $x/D = 9, 11, 14$ near the flame base are presented in Figs. 22–24, where each point represents filtered temperature-mixture fraction data from individual computational cells or from experiments. The upper solid curve denotes the equilibrium temperature for given mixture fractions under fast-reaction conditions, while the lower temperature boundary corresponds to the non-reacting mixing conditions. Experimental data and simulation results using both IEM and TCR mixing models illustrate the combined effects of SGS mixing inhomogeneity and finite-rate chemistry.

At $x/D = 9$ (Fig. 22), experimental measurements show minimal chemical influence with most points governed by large-scale flow mixing. IEM simulations exhibit pronounced combustion effects for mixture fractions below 0.8, while TCR results demonstrate combustion influence primarily in high-temperature regions along the jet periphery, aligning closer with experimental observations. However, in flow-dominated mixing regions (near the line connecting 1045 K at mixture fraction 0 and 305 K at mixture fraction 1), experimental data display scatter around the theoretical line that neither IEM nor TCR models reproduce. This discrepancy stems from both models' inability to approach Gaussian in isotropic turbulence without chemical reactions, which would require additional mixing-direction randomness absent in these formulations. The present study focuses on turbulence-chemistry interactions and does not address this specific limitation.

At $x/D = 11$ (Fig. 23), both experimental data and TCR model simulations exhibit triangular-distributed scatter points, capturing the coexistence of turbulence-dominated mixing and combustion effects ranging from weak to strong intensity. In contrast, IEM simulations demonstrate strong chemical influence, particularly for mixture fractions below 0.5, where nearly all points show combustion signatures and turbulence-controlled mixing points are virtually absent, which is a significant deviation from experiments. Similarly, at $x/D = 14$ (Fig. 24), while both models approximate experimental trends, TCR more accurately replicates the relative dominance of turbulent mixing over chemical reactions in high mixture fraction regions, exhibiting weaker chemical influence than IEM.

By analyzing Eq. (22) and IEM mixing constant reveals that TCR model produces weaker micro-mixing than IEM at low Da numbers (chemically weak regions), while potentially generating stronger micro-mixing at high Da numbers (chemically intense zones). Throughout the VCB, scalar fields experience dilution by combustion products and inert gases. Combined with the relationship between mixing homogeneity κ_α and filtered-to-PSR reaction rate ratio $\frac{\dot{\omega}_\alpha(\phi)}{\dot{\omega}_\alpha(\phi)}$ in Fig. 5, two regimes emerge: at low Da numbers (low κ_α region), small κ_α values mean increased micro-mixing (larger κ_α) elevates the reaction rate ratio $\frac{\dot{\omega}_\alpha(\phi)}{\dot{\omega}_\alpha(\phi)}$, enhancing chemical activity; conversely, at high Da numbers (high κ_α region), dilution effects cause increased homogeneity (larger κ_α) to reduce the filtered-to-PSR reaction rate ratio, suppressing reactions. Consequently, micro-mixing intensity modeled by TCR in VCB simulations may be either weaker or stronger than IEM. However, when κ_α approaches 0 or 1 under dilution, TCR consistently yields weaker reaction intensity than IEM, which is fully consistent with the behavior in scatter plots.

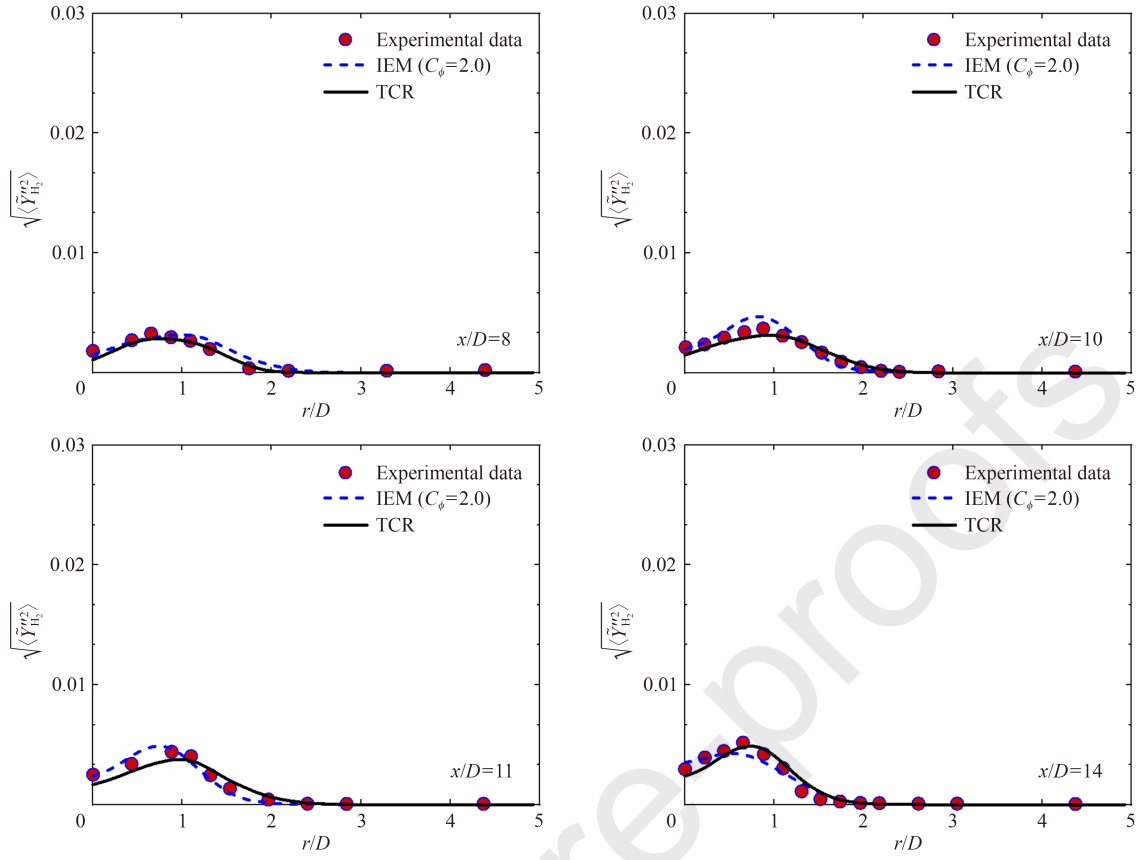


Fig. 19 Model predictions versus experimental measurements⁷³ of H_2 RMS fluctuations at multiple axial locations

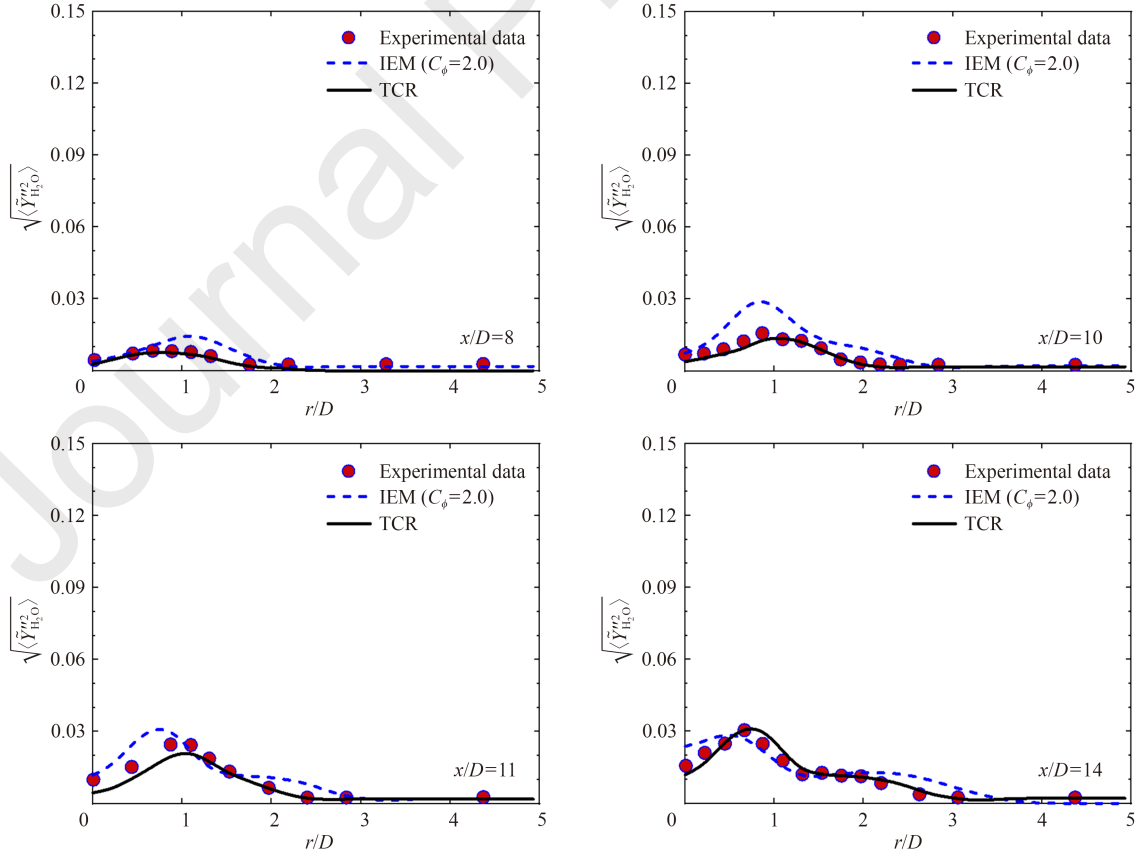


Fig. 20 Model predictions versus experimental measurements⁷³ of H_2O RMS fluctuations at multiple axial locations

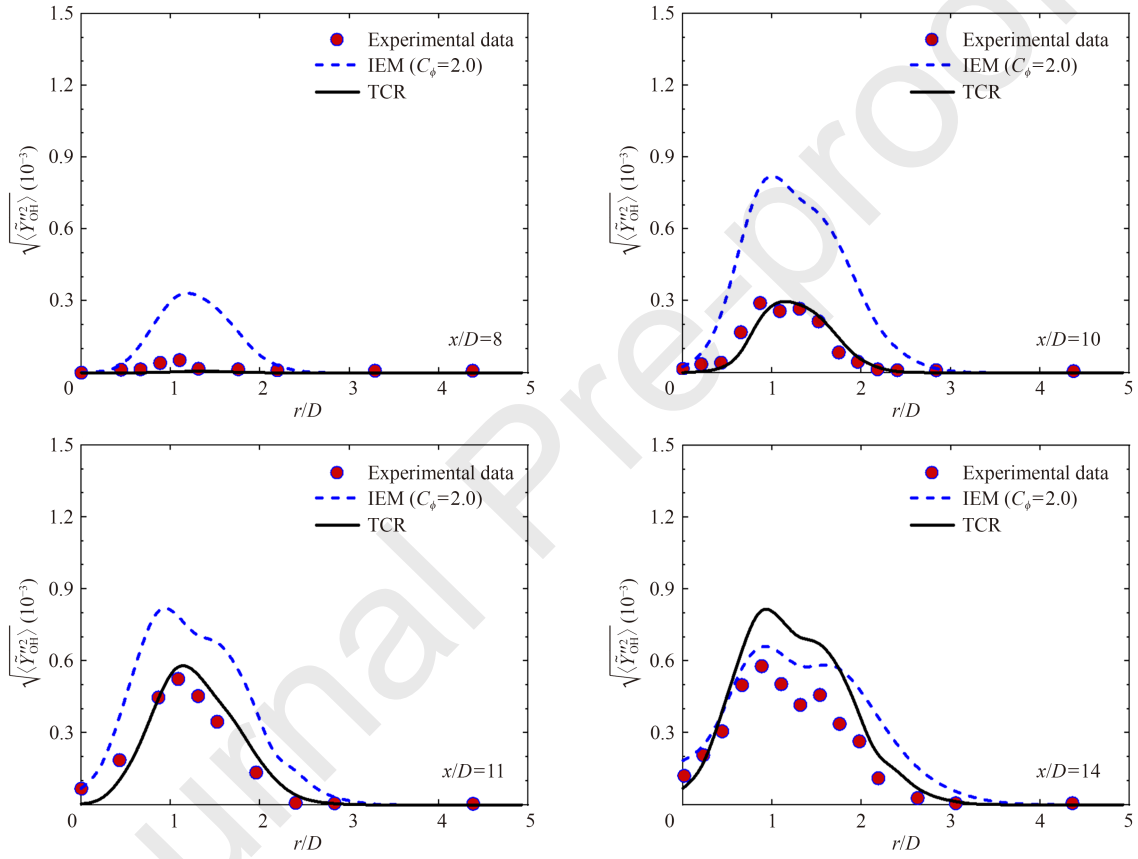


Fig. 21 Model predictions versus experimental measurements⁷³ of OH RMS fluctuations at multiple axial locations

Table 6 Radial-averaged relative error of RMS fluctuations of species mass fractions.

Species	Axial location	Radial-averaged relative error(%)	
		IEM	TCR
H ₂	$x/D = 8$	11.23	5.44
H ₂	$x/D = 10$	10.04	7.02
H ₂	$x/D = 11$	6.53	9.04
H ₂	$x/D = 14$	5.83	6.80
H ₂ O	$x/D = 8$	33.09	15.31
H ₂ O	$x/D = 10$	36.94	8.73
H ₂ O	$x/D = 11$	14.91	8.99
H ₂ O	$x/D = 14$	11.16	6.25
OH	$x/D = 8$	232.70	34.01
OH	$x/D = 10$	89.84	10.48
OH	$x/D = 11$	39.25	9.92
OH	$x/D = 14$	20.27	22.78

4.3. Effect of turbulence-chemistry interaction on micro-mixing

To investigate turbulence-chemistry interaction effects on micro-mixing, the TCR model simulations analyze the $H + O_2 \rightarrow O + OH$ reaction, treating O_2 as the scalar α in Eq. (22) to analyze its micro-mixing behavior.

Fig. 25 presents the ratio of micro-mixing terms computed by TCR versus IEM models in an identical transient combustion field. White regions indicate equivalent mixing intensity between models, red zones signify stronger TCR micro-mixing, and blue areas denote weaker TCR micro-mixing. Analysis reveals enhanced TCR

micro-mixing in chemically weak regions, while diminished relative to IEM in chemically intense zones. This behavior stems from strong chemistry sustaining larger SGS scalar gradients and higher variance, whereas weak chemistry promotes homogenization. IEM's micro-mixing exhibits weaker chemistry correlation, with its empirically tuned parameter $C_\phi = 2.0$ yielding mixing intensities between TCR's maximum and minimum values.

Fig. 26 presents a histogram distribution of κ_{O_2} values within strong reaction zones defined by OH mass fractions exceeding 0.00022, where bar height corresponds to the grid cell count in each κ_{O_2} interval. The distribution reveals two dominant regimes with a substantial portion of grid cells exhibiting κ_{O_2} values in $0.9 < \kappa_{O_2} \leq 1$, indicating near-complete mixing where the micro-mixing target closely approximates the local filtered O_2 mass fraction, while significant cell counts populate the range $0 < \kappa_{O_2} \leq 0.3$. Considering the dynamically determined $C_\phi \approx 10$ for weak-reaction conditions, TCR mixing intensity at $\kappa_{O_2} \approx 0.2$ becomes equivalent to IEM with $C_\phi = 2$, while $\kappa_{O_2} < 0.2$ yields weaker micro-mixing than conventional IEM implementations.

Fig. 27 presents the correlation between filtered-to-PSR reaction rate ratio $\frac{\overline{\dot{\omega}_{O_2}(\phi)}}{\dot{\omega}_{O_2}(\phi)}$ and mixing homogeneity κ_{O_2} across the computational domain in TCR simulations, where each scatter point represents a grid cell with non-zero reaction rate. All η_{O_2} values remain below 0.5. For fixed η_{O_2} , the filtered-to-PSR ratio initially increases with κ_{O_2} , reaching a maximum exceeding unity before gradually decreasing. This non-monotonic behavior aligns with theoretical predictions from Eq. (19) and shows excellent agreement with the reference curve in Fig. 5.

The scatter plot in Fig. 27 illustrates the general relationship between $\frac{\overline{\dot{\omega}_{O_2}(\phi)}}{\dot{\omega}_{O_2}(\phi)}$ and κ_{O_2} but cannot quantify κ_{O_2} distribution due to point overlap. Analysis of κ_{O_2} frequency distribution in Fig. 26 reveals predominant values near 0 and 1, indicating that for most grid cells, the filtered-to-PSR reaction rate ratio changes substantially with mixing homogeneity.

At $\kappa_{O_2} \approx 0$, the filtered-to-PSR reaction rate ratio increases significantly with κ_{O_2} where dilution effects are weak and homogeneous mixing promotes reactions. Since TCR produces weaker mixing than IEM at low κ_{O_2} , it yields lower filtered reaction rates

at these locations. Conversely at $\kappa_{O_2} \approx 1$ where $\frac{\overline{\dot{\omega}_{O_2}(\phi)}}{\dot{\omega}_{O_2}(\phi)} \geq 1$, the filtered-to-PSR reaction rate ratio decreases markedly with increasing κ_{O_2} as enhanced homogenization intensifies dilution. TCR generates stronger mixing than IEM at high κ_{O_2} , again resulting in lower filtered reaction rates. Intermediate κ_{O_2} values ($0.3 < \kappa_{O_2} \leq 0.8$) exhibit minimal ratio variation with κ_{O_2} , being primarily governed by bulk reactant concentrations. Fig. 26 confirms these intermediate values constitute a minority of cells.

Collectively, the SGS mixing state represented by the TCR model creates less favorable conditions for reactions compared to the IEM model, explaining the elevated flame lift-off height observed in TCR simulations. This finding aligns with the superior agreement between lift-off heights and combustion fields predicted by TCR and VCB experimental data, indicating systematic overestimation of filtered reaction rates by IEM in most reaction zones. The physically consistent behavior of TCR, which demonstrates weaker mixing than IEM in high Da number regions and stronger mixing in low Da number regions, establishes improved realism in turbulent combustion modeling.

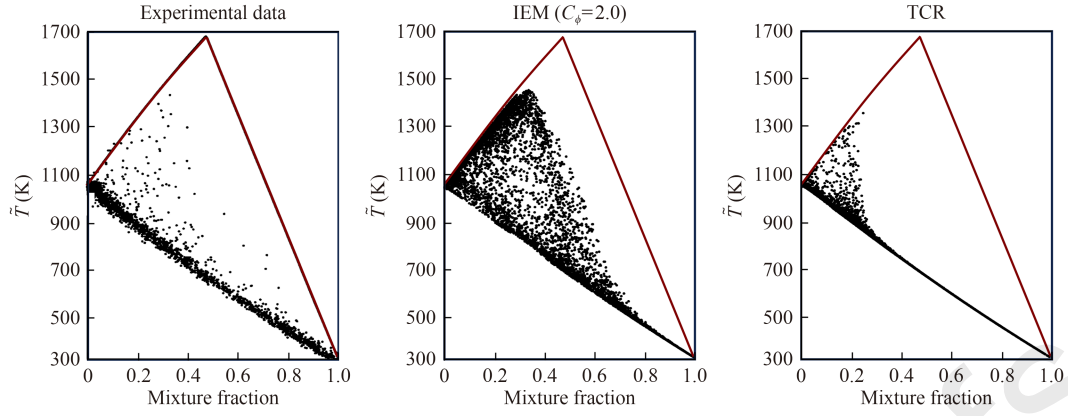


Fig. 22 Scatter plots of temperature at $x/D = 9$ near flame base: experimental (left), computed using IEM (middle) and TCR (right) mixing model.

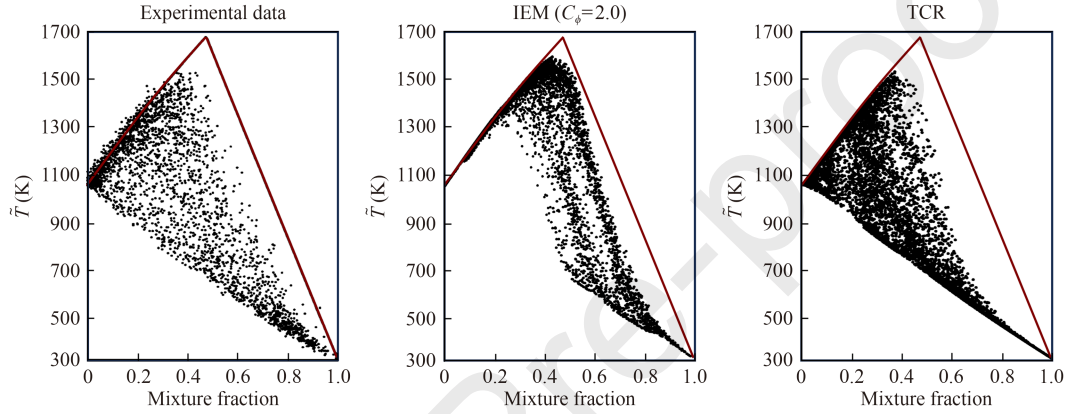


Fig. 23 Scatter plots of temperature at $x/D = 11$ near flame base: experimental (left), computed using IEM (middle) and TCR (right) mixing model.

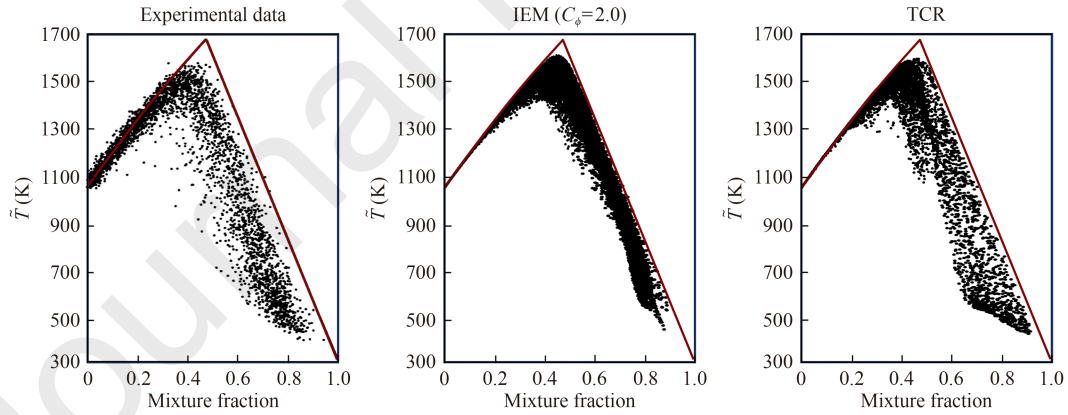


Fig. 24 Scatter plots of temperature at $x/D = 14$ near flame base: experimental (left), computed using IEM (middle) and TCR (right) mixing model.

5. Conclusions

This study proposes a Turbulence-Chemistry Recursive (TCR) mixing model that drives scalar fields toward dynamic equilibrium states achieved through turbulence-chemistry interactions, rather than conventional homogenization. The model establishes a self-consistent framework which: first, quantifies chemical inhomogeneity via the relationship between filtered reaction rates from TPDF solutions and PSR reference values; second, employs a novel multi-zone partitioning method extending PaSR concepts to quantify SGS

scalar distribution inhomogeneity as ψ_α^n ; and finally applies the same partitioning approach to model micro-mixing and stochastic variable variance decay rates within the control volume, recursively incorporating this into the TPDF framework's micro-mixing process. This approach directly utilizes TPDF solutions to quantify scalar inhomogeneity, eliminating ambiguous timescale modeling and parameter tuning inherent in conventional PaSR methods. It accommodates a wide Damköhler number range: at low Da (resembling broken reaction zones), the control volume approaches a PSR state where TCR replicates IEM's rapid variance decay;

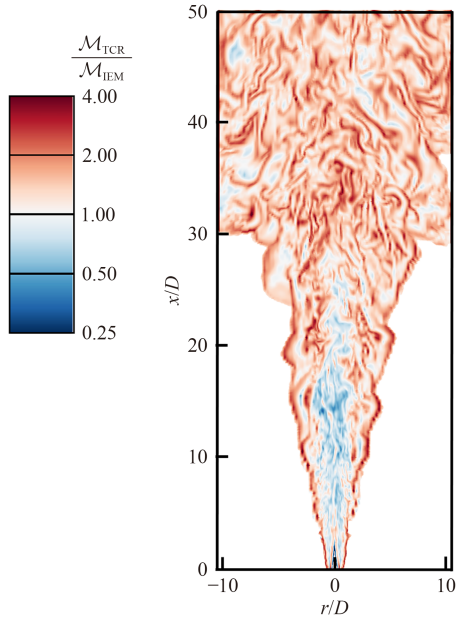


Fig. 25 Relative transient micro-mixing intensity of TCR versus IEM model.

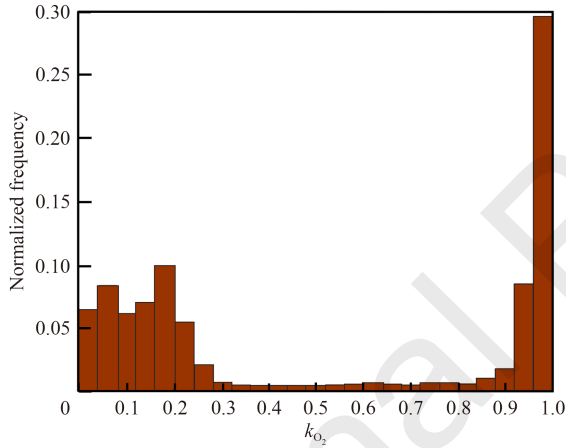


Fig. 26 Normalized frequency distribution of κ_{O_2} in reaction zones ($Y_{OH} \geq 0.00022$).

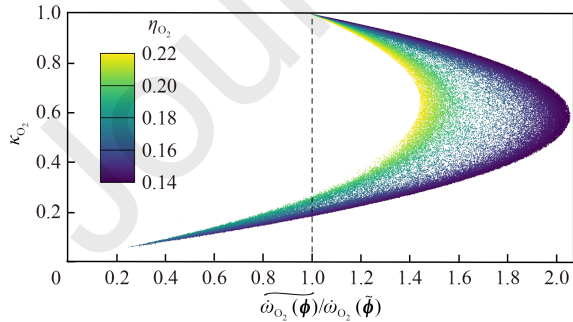


Fig. 27 Scatter plot of mixing homogeneity κ_{O_2} versus filtered-to-PSR reaction rate ratio in a transient combustion field.

at high Da (flamelet-like regimes), the PSR fraction nears zero, preserving variance. By proportionally combining these states, TCR reflects inhomogeneity quantification in stochastic variable sampling, achieving broad-range accuracy while maintaining IEM-

comparable computational efficiency and ESF compatibility.

The TCR model retains IEM's fundamental mathematical properties of mass conservation, variance decay, boundedness, linearity, and independence, while introducing three critical modifications: it incorporates local chemical inhomogeneity effects through κ_α -based reaction rate disparities for improved SGS variance prediction; establishes ξ_α^n -relative micro-mixing targets enabling localness; and generates stochastic variability via TPDF-derived reaction rates while maintaining performance in inhomogeneous flows despite non-Gaussian relaxation in non-reacting regions. In flamelet regimes where reactions strongly enhance scalar gradients, TCR naturally reduces mixing intensity ($\kappa_\alpha \rightarrow 0$) to inherently capture reaction-driven gradient steepening without explicit dissipation adjustments. The TCR model quantifies chemical inhomogeneity through the relationship between filtered reaction rates derived from TPDF solutions and PSR reaction rates. This quantified inhomogeneity serves to measure SGS scalar variance, which is then recursively incorporated into the micro-mixing process within the TPDF framework. This recursive integration provides a more theoretically grounded approach than methods that directly modify mixing timescales using turbulence-chemistry timescales³³.

Validation in turbulent lifted flames demonstrates superior predictive capability of the TCR model relative to conventional IEM closure. Simulations of the VCB within the ESF-TPDF framework confirm that TCR accurately captures flame lift-off height and combustion structure, whereas IEM erroneously predicts premature ignition and excessive reaction rates in the flame base region. Quantitative analyses reveal that TCR reduces mean temperature prediction errors by 76% (from 10.20% to 2.46%) and temperature RMS errors by 61% (from 20.65% to 8.08%) compared to IEM. For hydroxyl radical concentrations, TCR achieves an order-of-magnitude error reduction, lowering the average error from 179.57% to 16.81% while substantially improving RMS fluctuation predictions. These results confirm autonomous adaptation of TCR to turbulence-chemistry interactions in lifted flame configurations.

Scatter analyses at upstream locations show the TCR model better aligns with experimental temperature-mixture fraction distributions, accurately capturing limited combustion influence and turbulence-combustion coexistence while IEM exhibits premature reactions and excessive chemical dominance. This originates from autonomous micro-mixing modulation in the TCR model, producing weaker mixing than IEM in low-Damköhler number regions near $\kappa_\alpha = 0$ yet stronger mixing in high-Damköhler zones near $\kappa_\alpha = 1$. Under vitiated coflow dilution conditions, TCR consistently yields lower filtered reaction rates than IEM at both mixing extremes. Frequency distribution analyses confirm that κ_α values predominantly occupy near-zero reacting states or near-unity non-reacting states, with negligible occurrence of intermediate homogeneity states. This distribution explains systematic suppression of chemical reactivity by TCR relative to IEM. The resulting elevated flame lift-off heights align with experimental observations, demonstrating improved capacity of the TCR model to capture finite-rate chemistry effects in turbulent lifted flames.

Given these results, the TCR model establishes a recursive closure framework by quantifying scalar inhomogeneity arising from turbulence-chemistry interactions to formulate the micro-mixing model. This empirical-constant-free approach incorporates chemical effects in micro-mixing, enabling simultaneous capture of turbulence-driven homogenization and chemistry-driven differentiation. Validation in turbulent lifted flames demonstrates superior accuracy. Future developments should implement enhanced mixing-direction randomness to ensure Gaussian convergence in isotropic

non-reactive turbulence, with extended validation across diverse combustor configurations and combustion scenarios, including Direct Numerical Simulation (DNS)-based verification of chemistry-induced micro-mixing mechanisms.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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